

Source: US20230420457A1 | Corresponding grant: US12568683B2

FIG. 2A

100

200

X

X₂/X₁

PPT CROSS GATE

FIG. 10

100
G

YD+1
YD+2

CROSS FINGER GATE REGION

CROSS FINGER GATE REGION

200

200

FIG. 3B 100
 $\hat{z}$

$\hat{z}$
 $\hat{y}$ $\hat{x}$

Y1-Y1
CROSS FIN GATE REGION

Y2-Y2
CROSS FIN SG REGION

The figure shows two cross-sectional views of a device. The left view, labeled 'Y1-Y1 CROSS FIN GATE REGION', shows a substrate 200 with a gate stack 300 on top. The gate stack 300 consists of a bottom layer 302 and a top layer 304. The right view, labeled 'Y2-Y2 CROSS FIN SG REGION', shows a similar structure with a substrate 200 and a gate stack 300. The gate stack 300 consists of a bottom layer 302 and a top layer 304. A coordinate system is shown in the top left with axes $\hat{x}$, $\hat{y}$, and $\hat{z}$.

Figure 4 consists of two schematic diagrams, (a) and (b), showing the cross-sections of two types of cross-gates. Diagram (a) is labeled 'XG-81 PRECAST CROSS-GATE' and diagram (b) is labeled 'XG-82 CAST-IN-PLACE CROSS-GATE'. Both diagrams show a central vertical section with two side sections. The central section is labeled '100' at the top and '100' at the bottom. The side sections are labeled '100' at the top and '100' at the bottom. The diagrams show the internal structure of the gates, including the concrete and the reinforcement. The reinforcement is shown as horizontal lines within the concrete sections. The diagrams are labeled with 'a)' and 'b)' at the bottom.

[illegible]

FIG. 2D **2000**
2

FORM A PFET COMPRISING AT LEAST ONE SILICON GERMANIUM CHANNEL. 2000

3

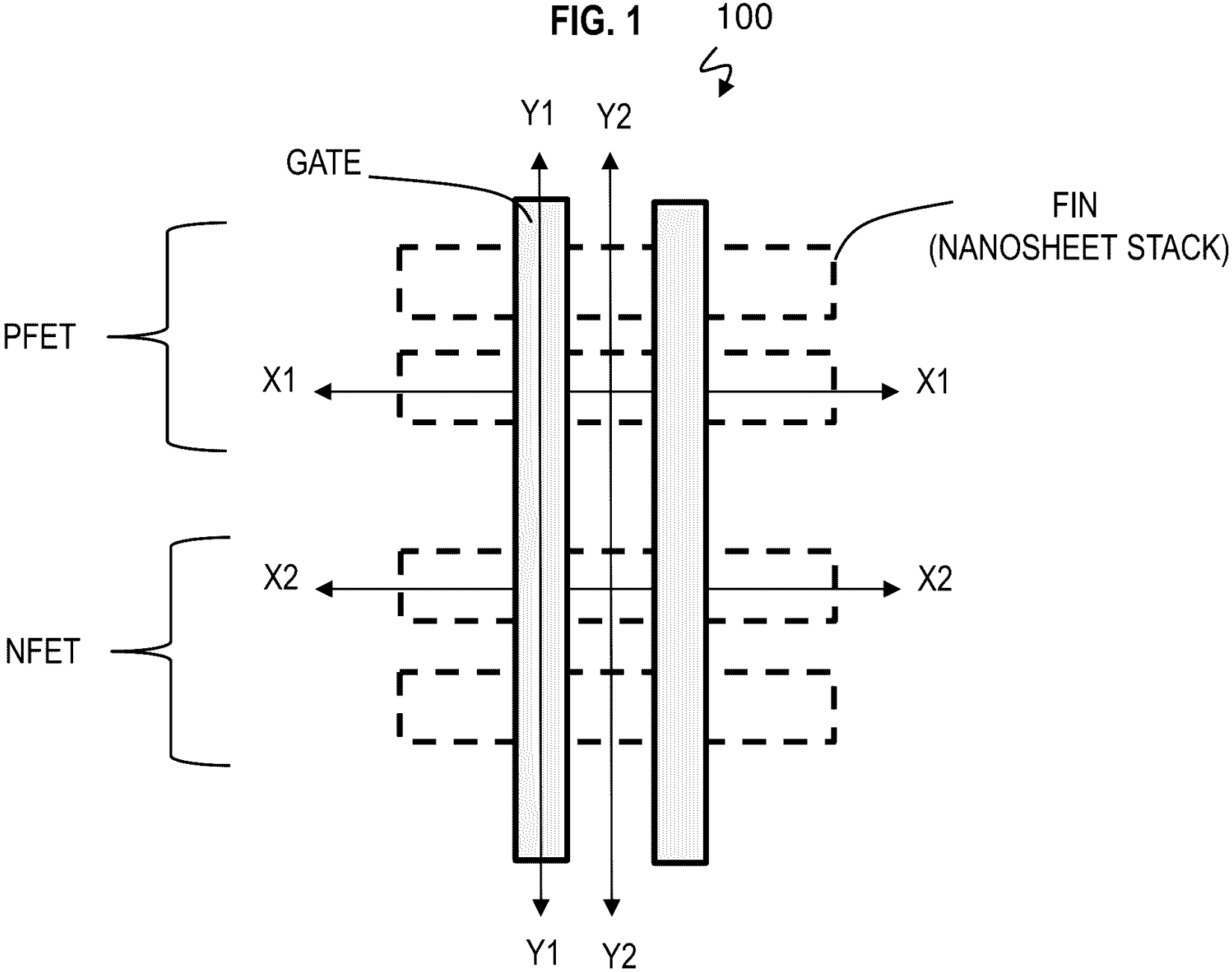
FORM A PFET COMPRISING AT LEAST ONE SILICON CHANNEL. THE PFET BEING POSITIONED LATEROALLY TO THE MPET, THE AT LEAST ONE SILICON CHANNEL AND THE AT LEAST ONE SILICON GERMANIUM CHANNEL BEING ENGAGED IN A VERTICAL DIRECTION. 2000

FIG. 21 $\frac{2100}{2}$

FORM A FIRST NANOWEETH STACK AND A SECOND NANOWEETH STACK. EACH OF THE FIRST AND SECOND NANOWEETH STACKS COMPRISING A FIRST LAYER, SECOND LAYERS, AND THIRD LAYERS 1110
↓
REPLACE THE FIRST LAYER IN THE SECOND NANOWEETH STACK WITH A BOTTOM ISOLATION LAYER 1100
↓
REPLACE THE FIRST LAYER AND THE THIRD LAYERS IN THE FIRST NANOWEETH STACK WITH FIRST GUIDE STRIP MATERIAL 1100
↓
REPLACE THE SECOND LAYERS IN THE SECOND NANOWEETH STACK WITH SECOND GUIDE STRIP MATERIAL 1100

<https://patents.google.com/patent/US20230420457A1/en>

Figure 1



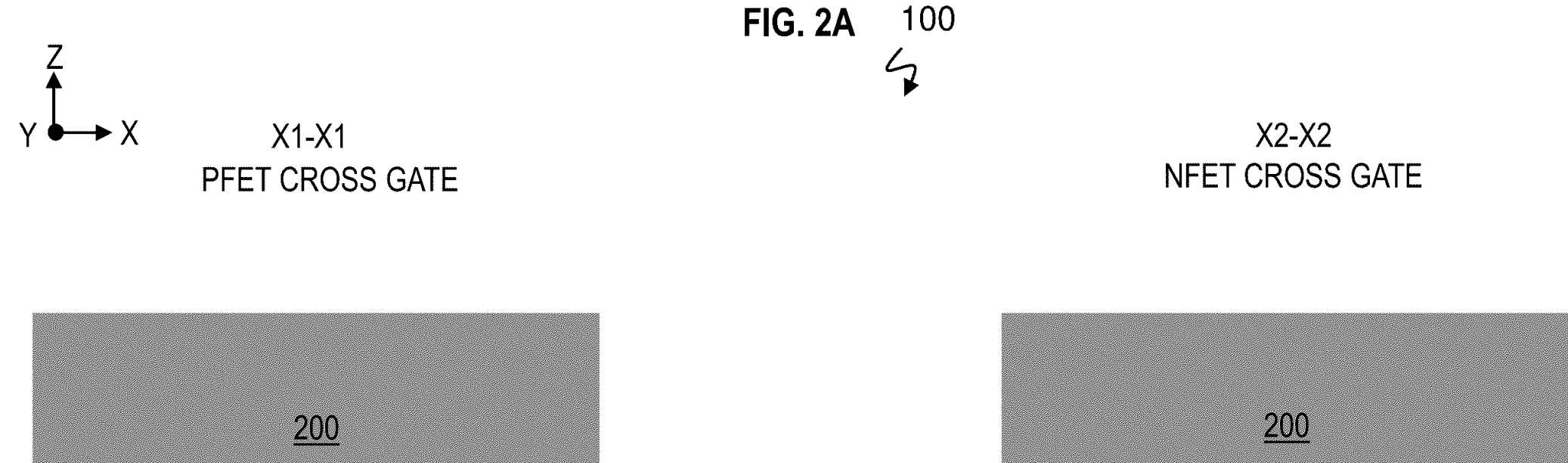
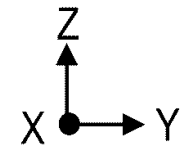
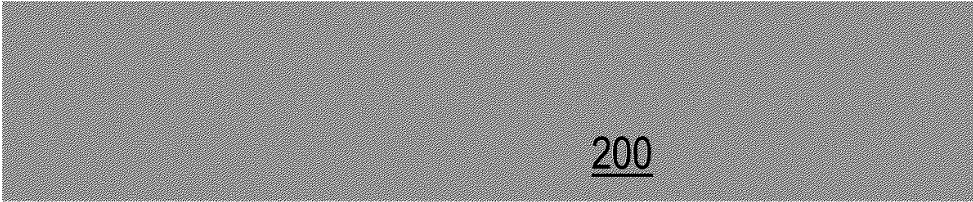
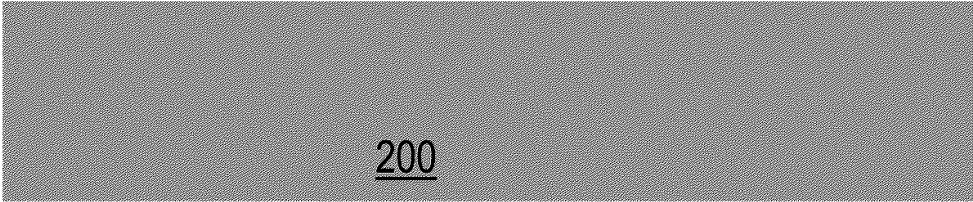


FIG. 2B 100



Y1-Y1
CROSS FIN GATE REGION

Y2-Y2
CROSS FIN S/D REGION



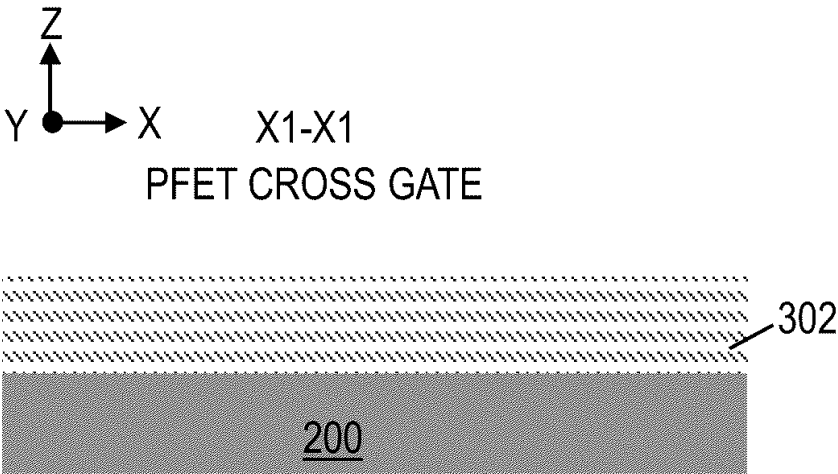


FIG. 3A 100

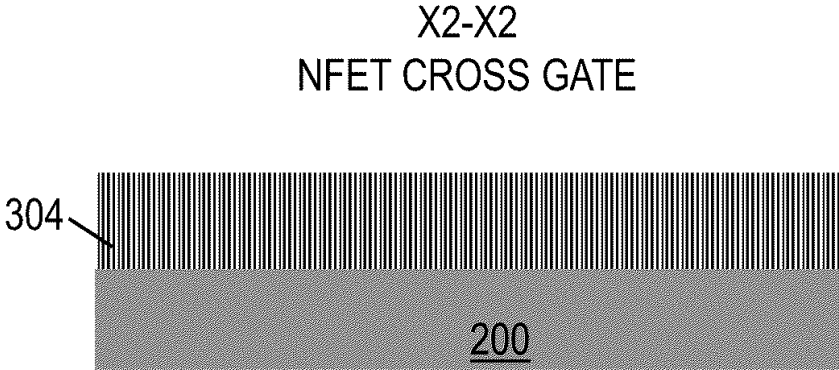
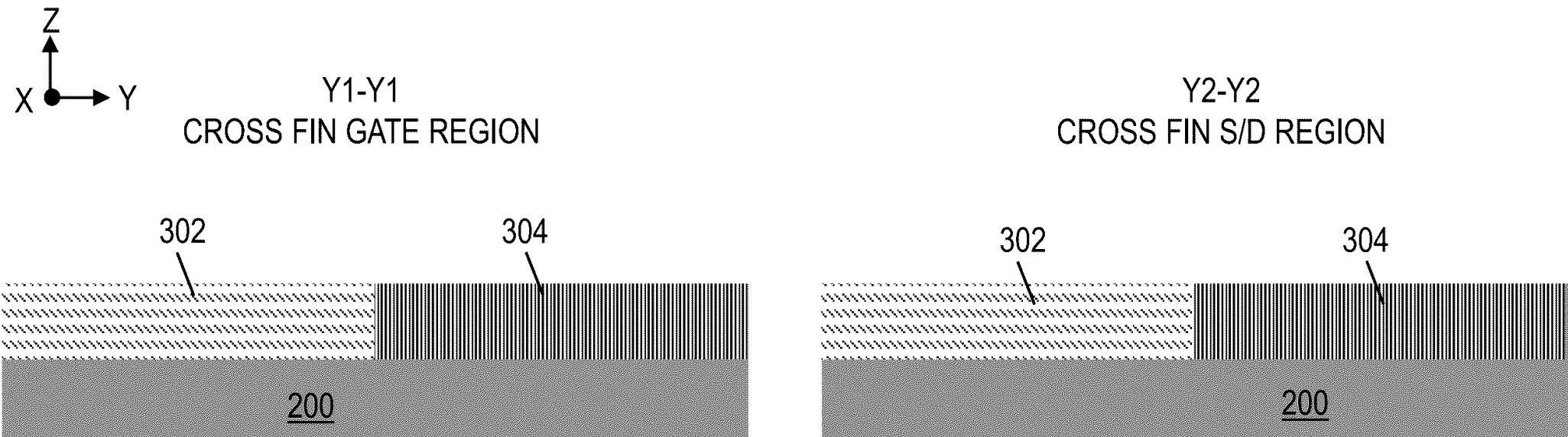
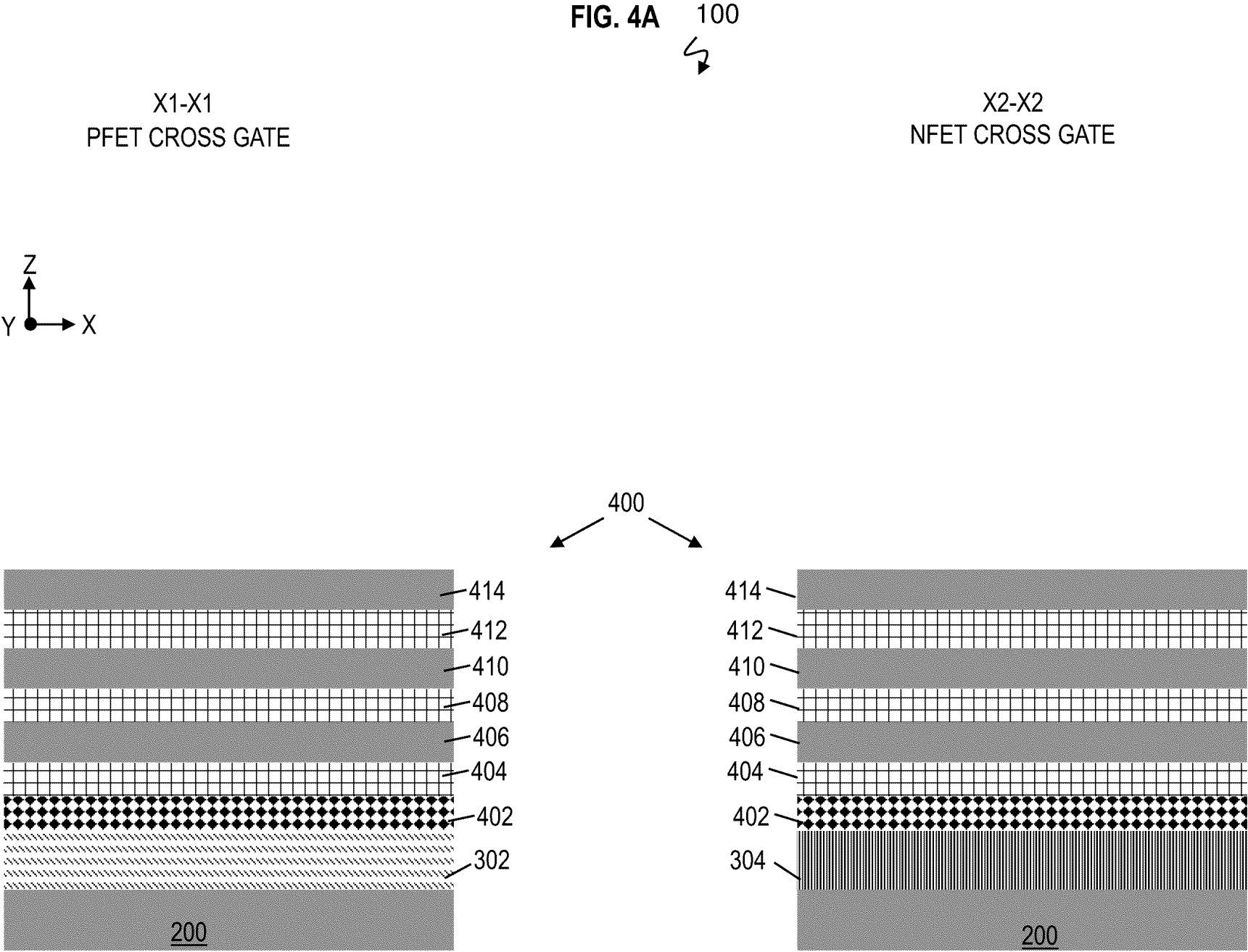
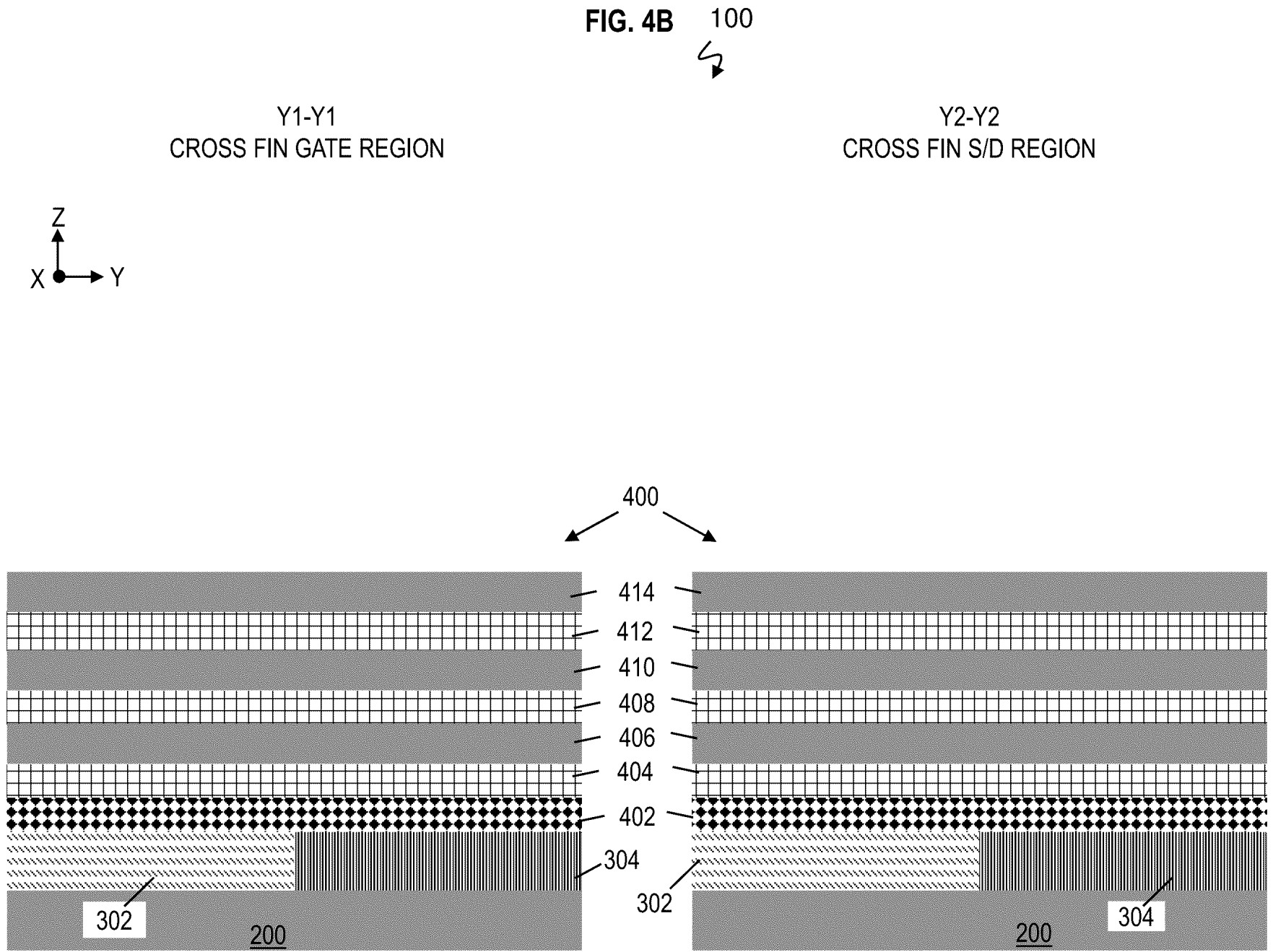


FIG. 3B 100







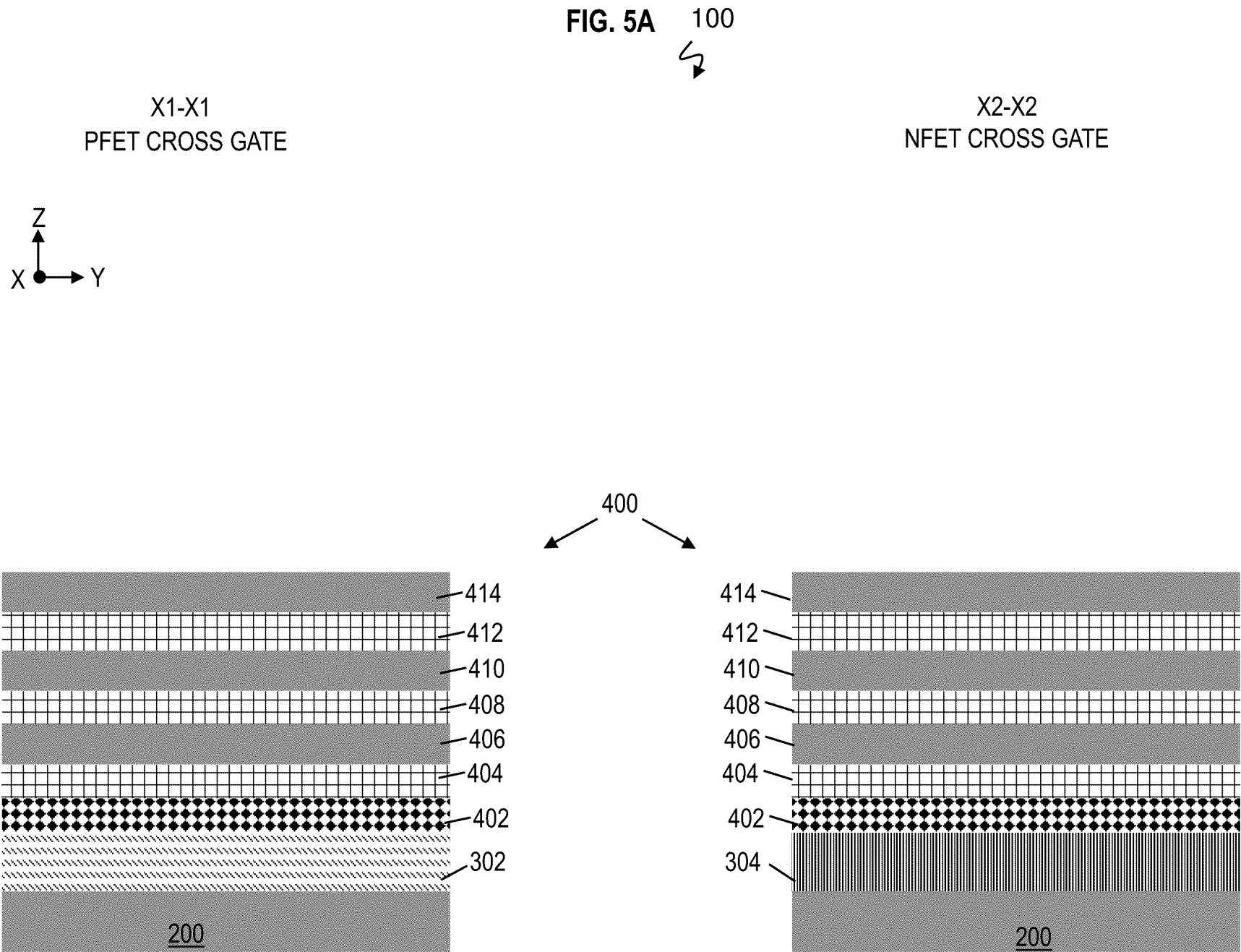
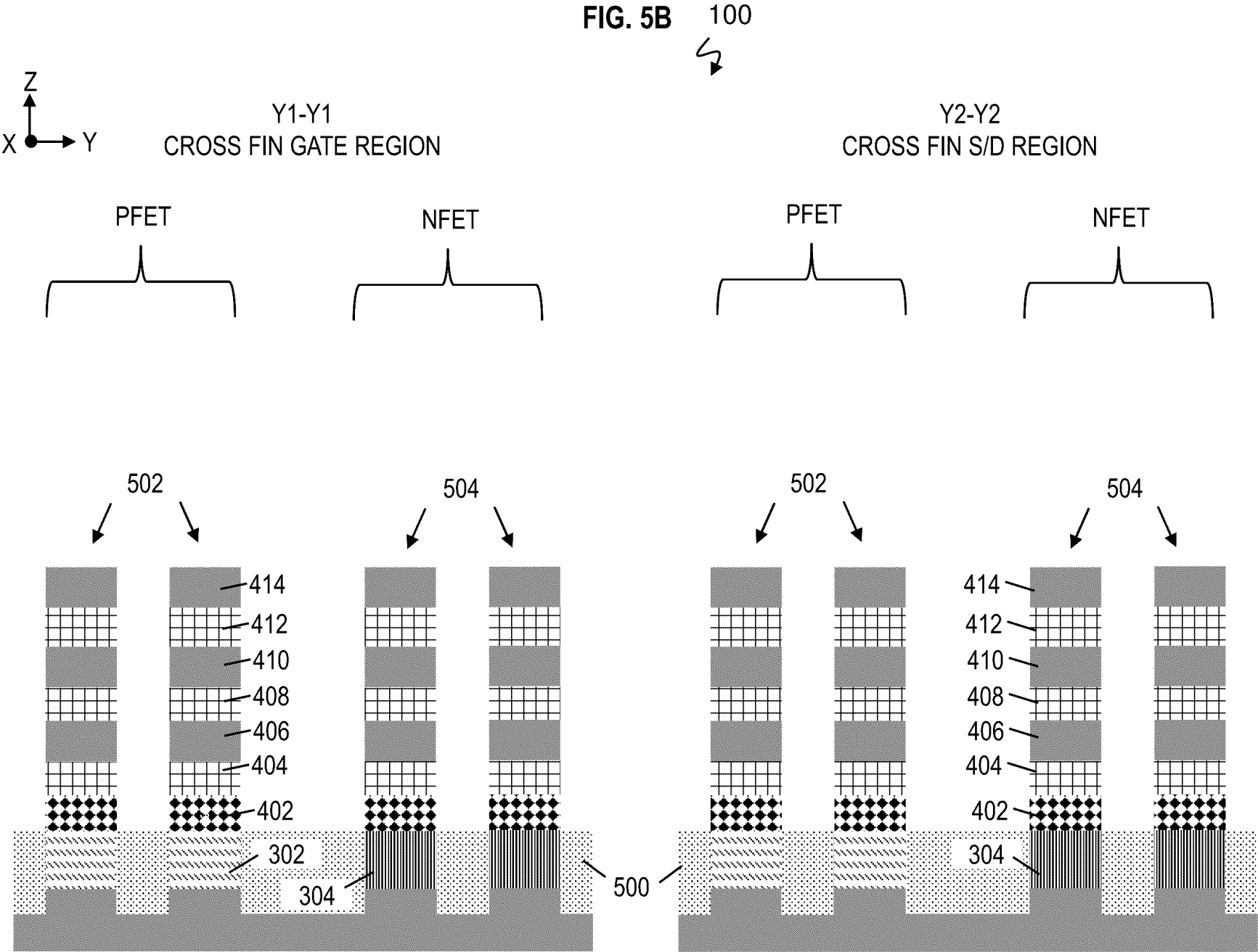


Figure 5B



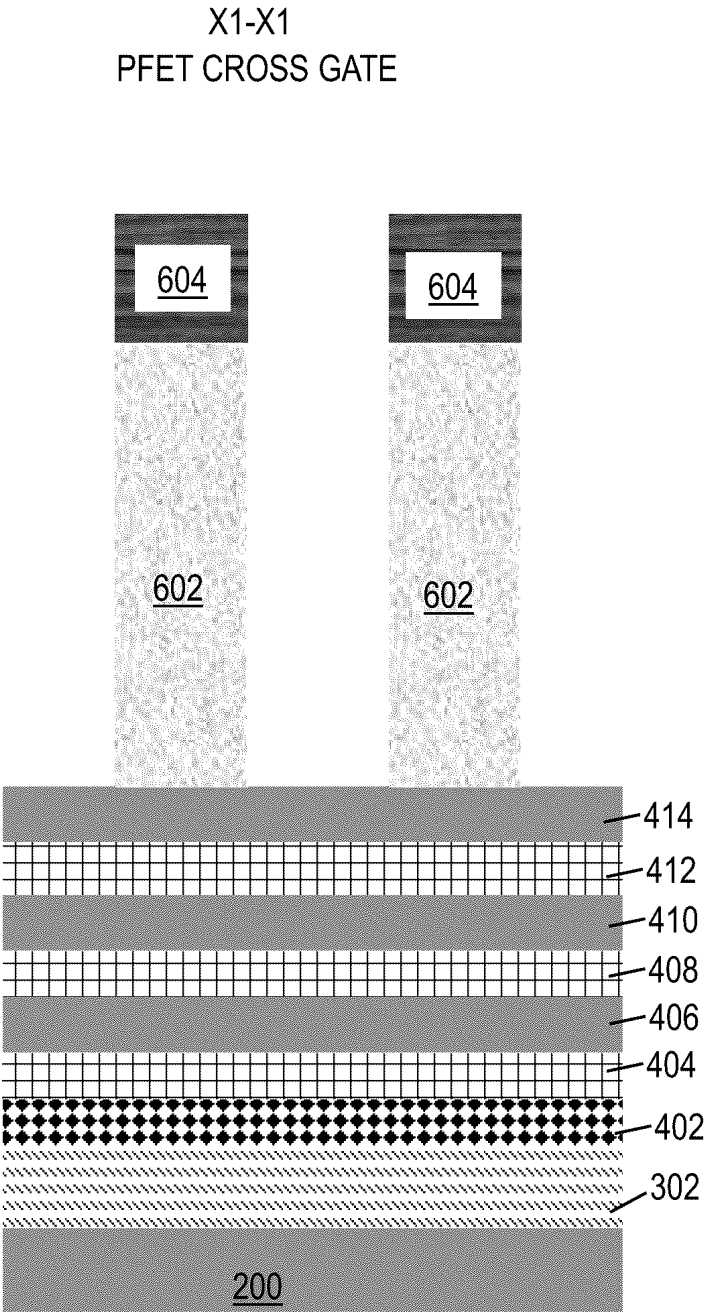


FIG. 6A 100

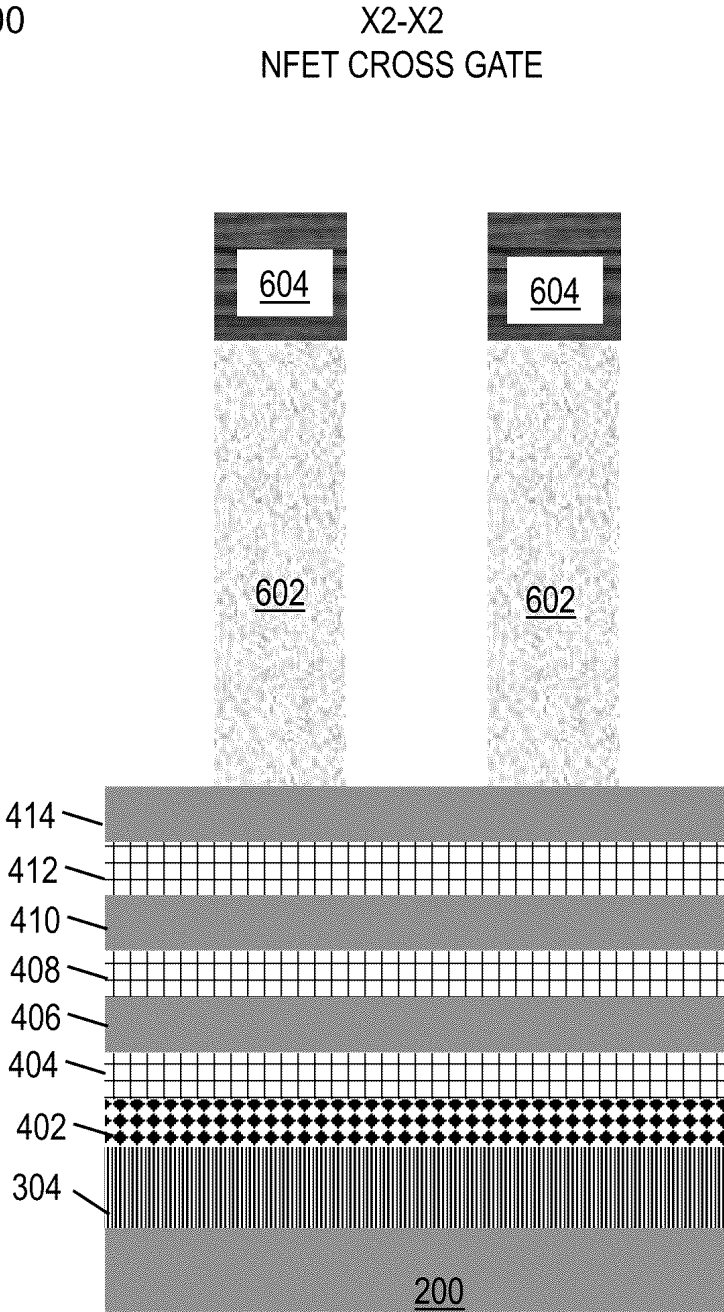




Figure 7A

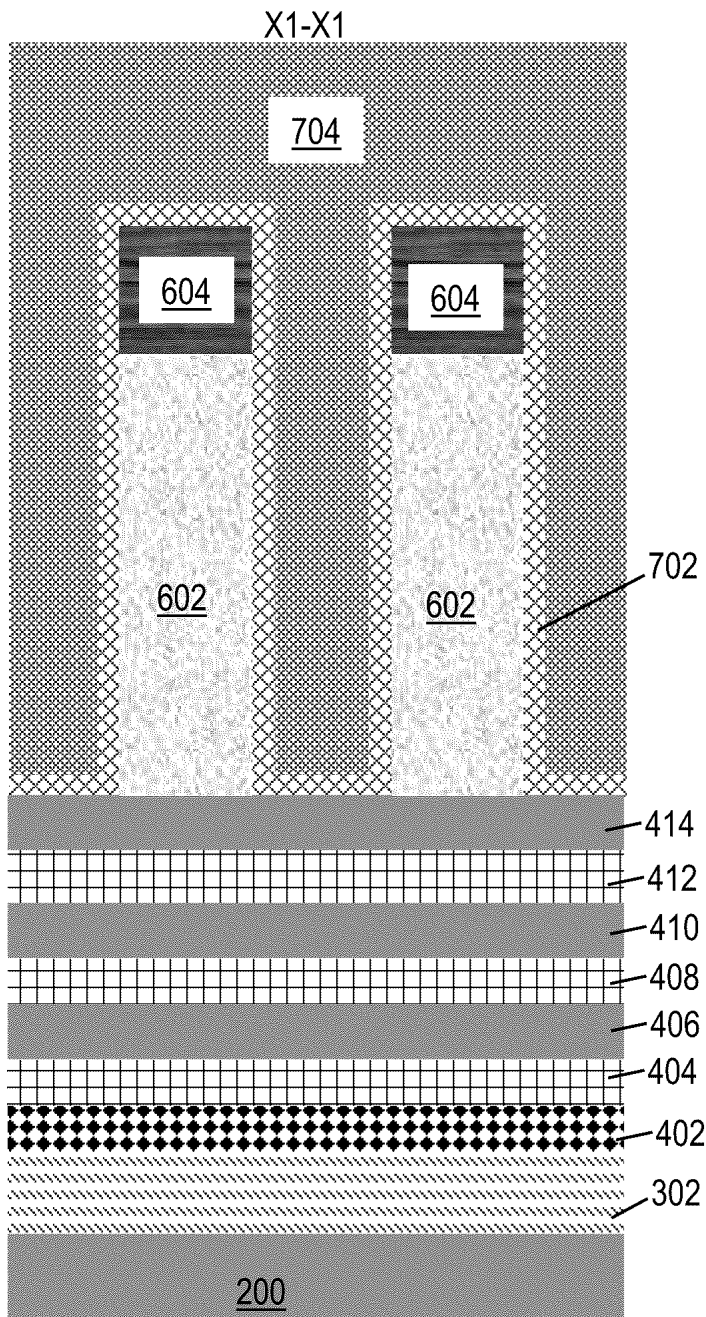


FIG. 7A 100

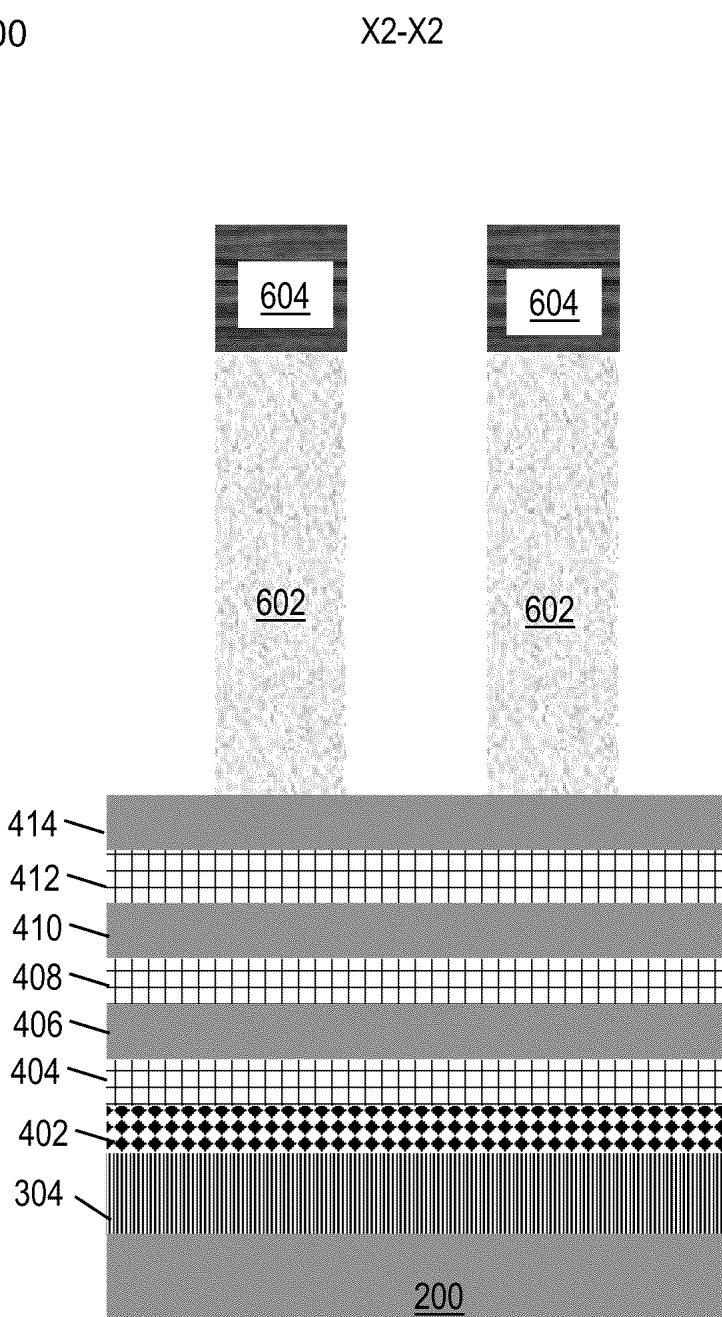
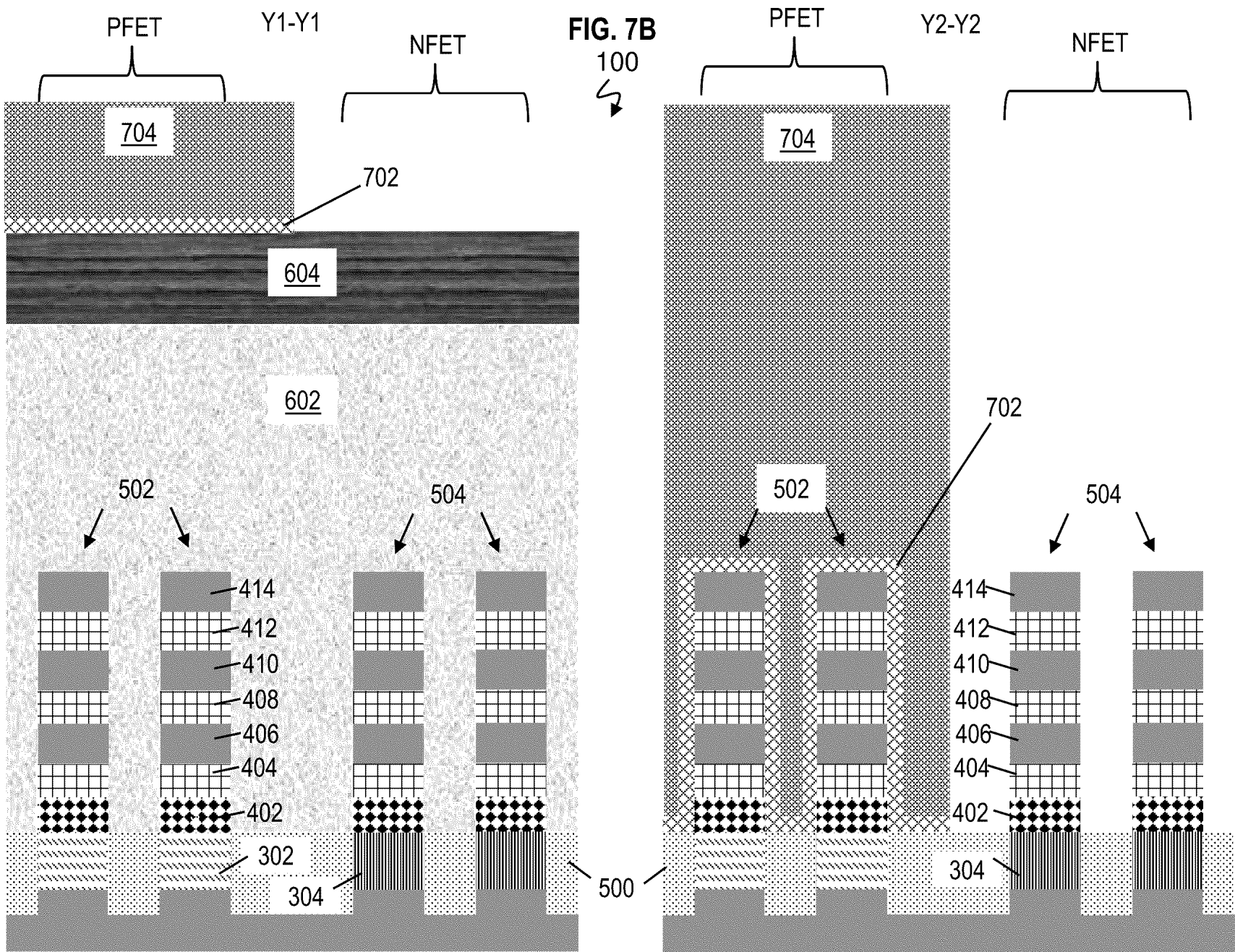
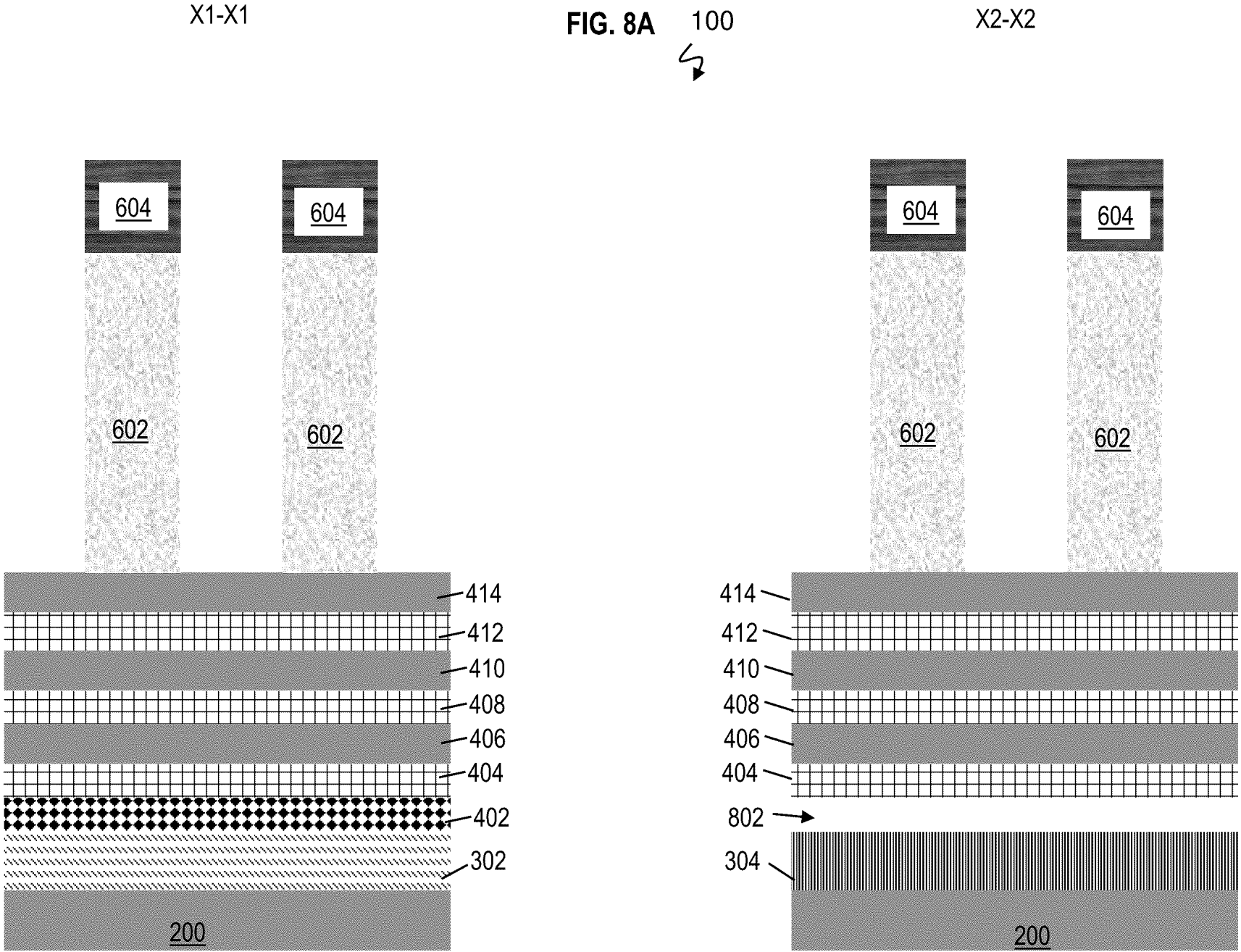
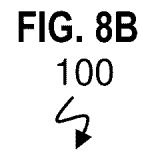


Figure 7B







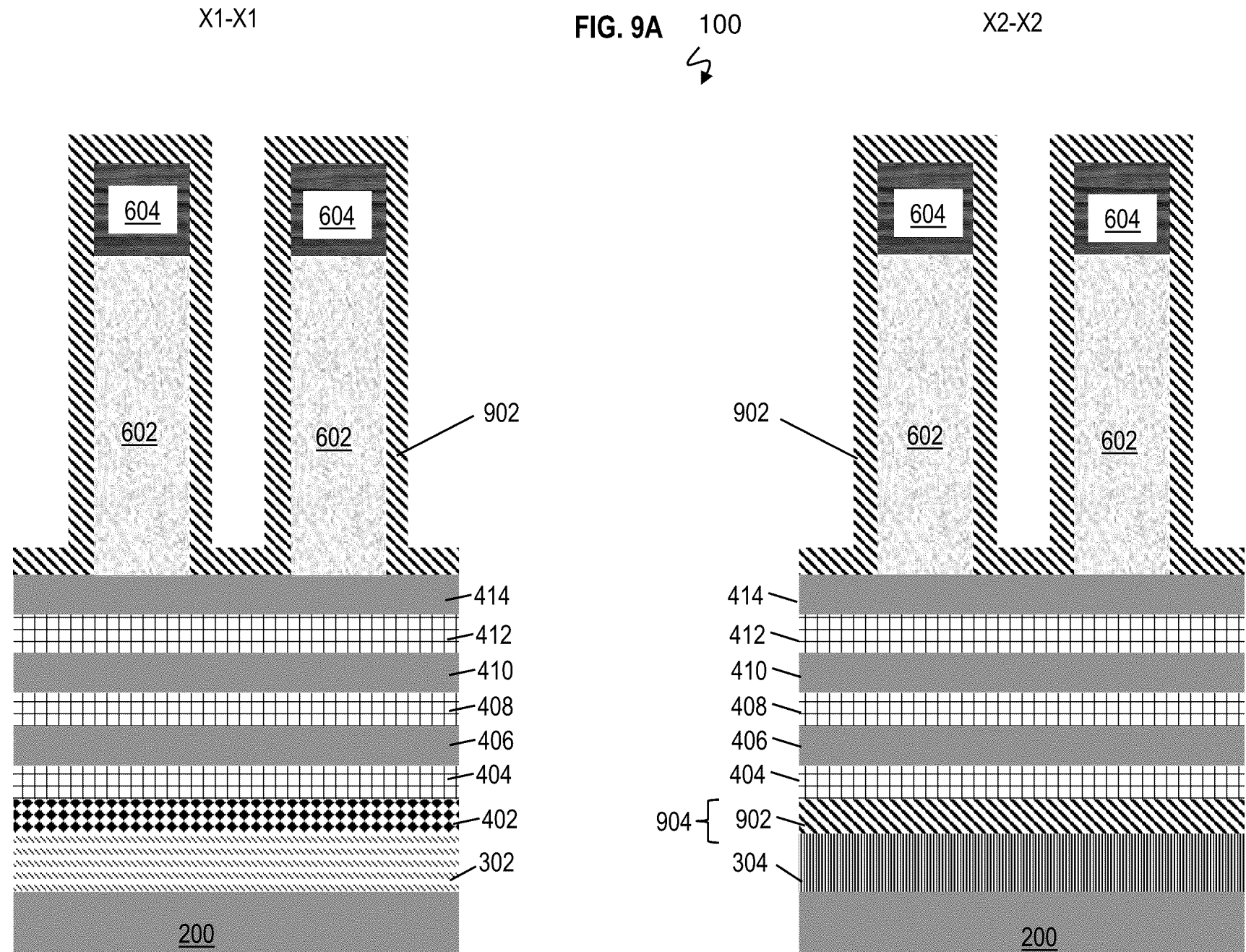
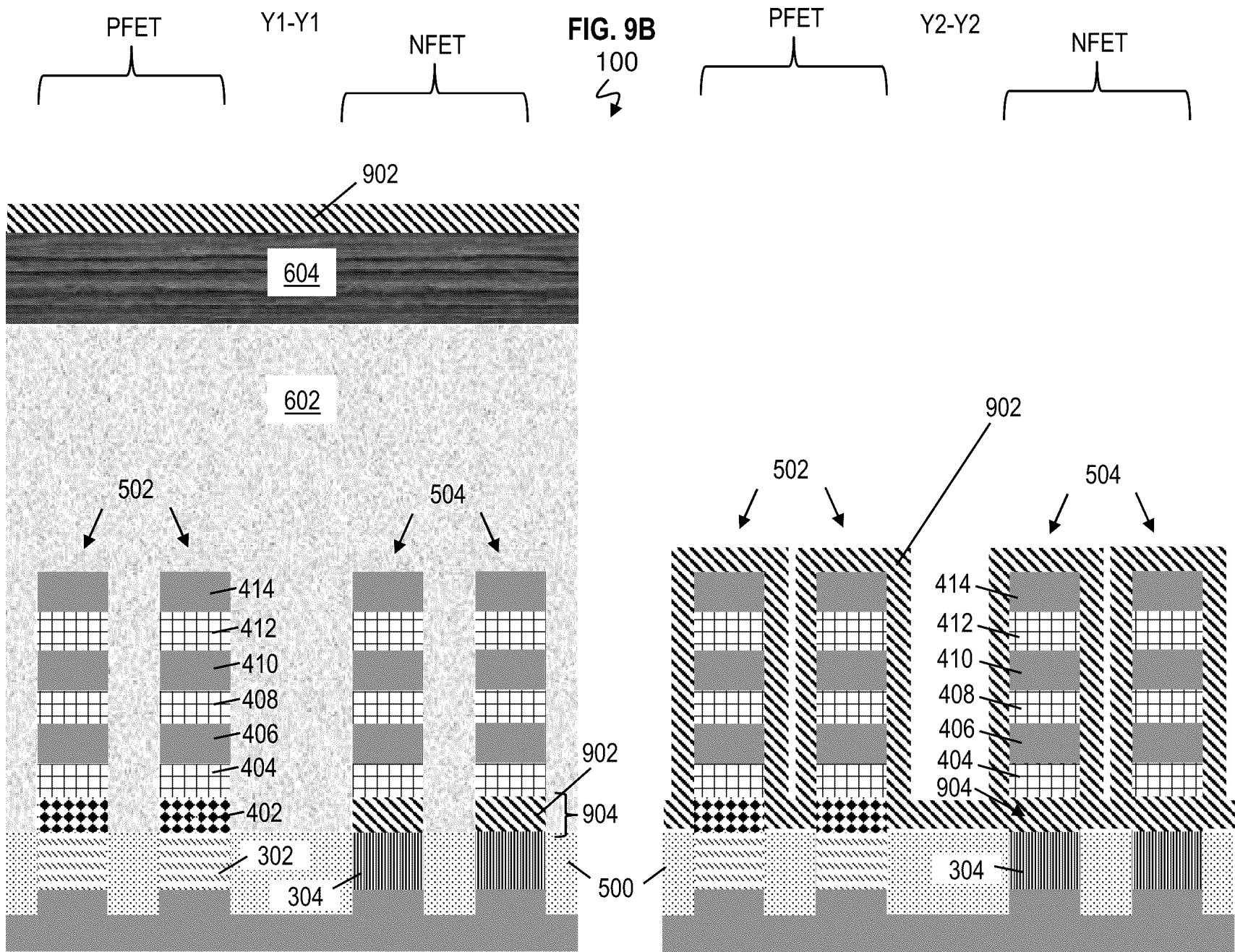


Figure 9B



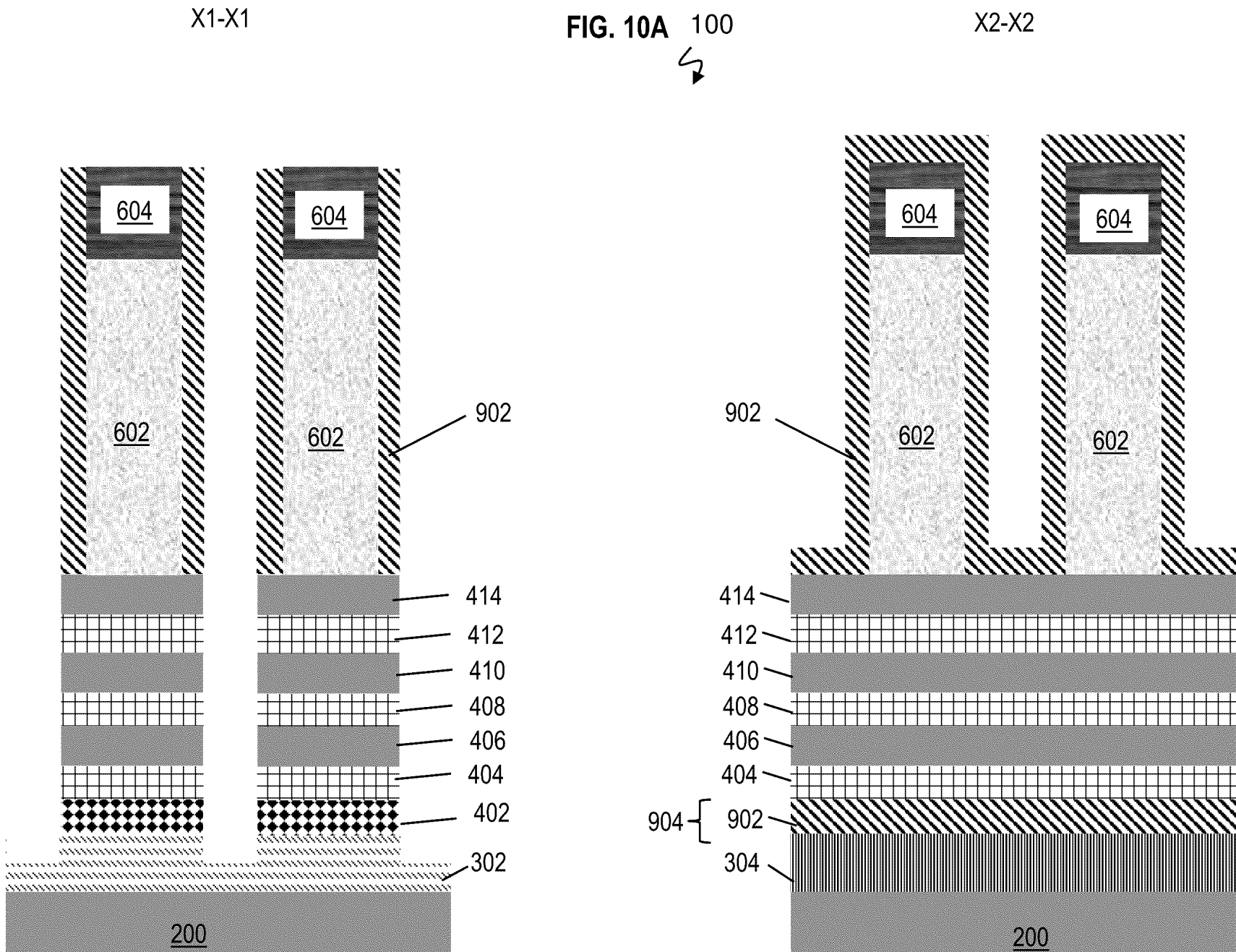
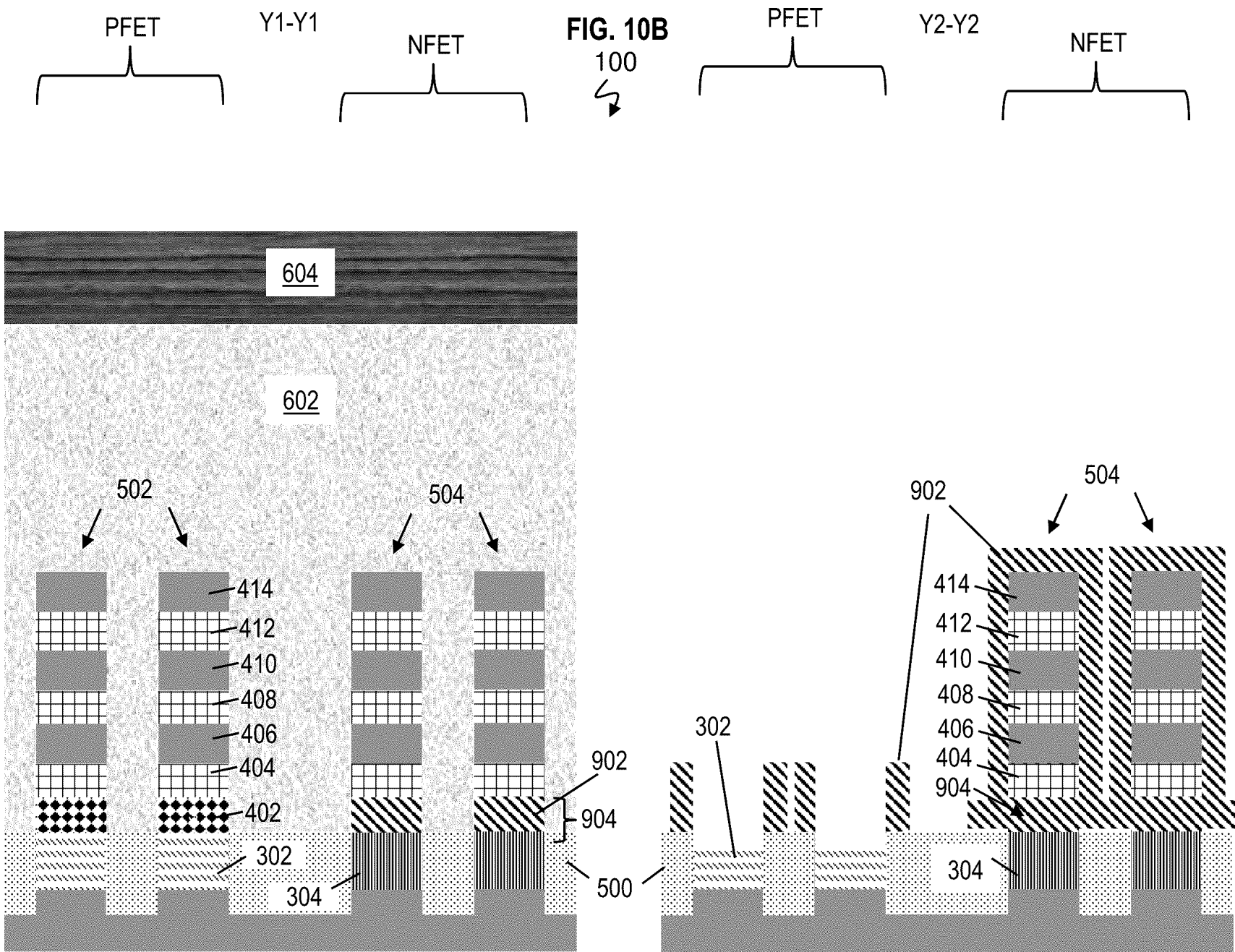


Figure 10B



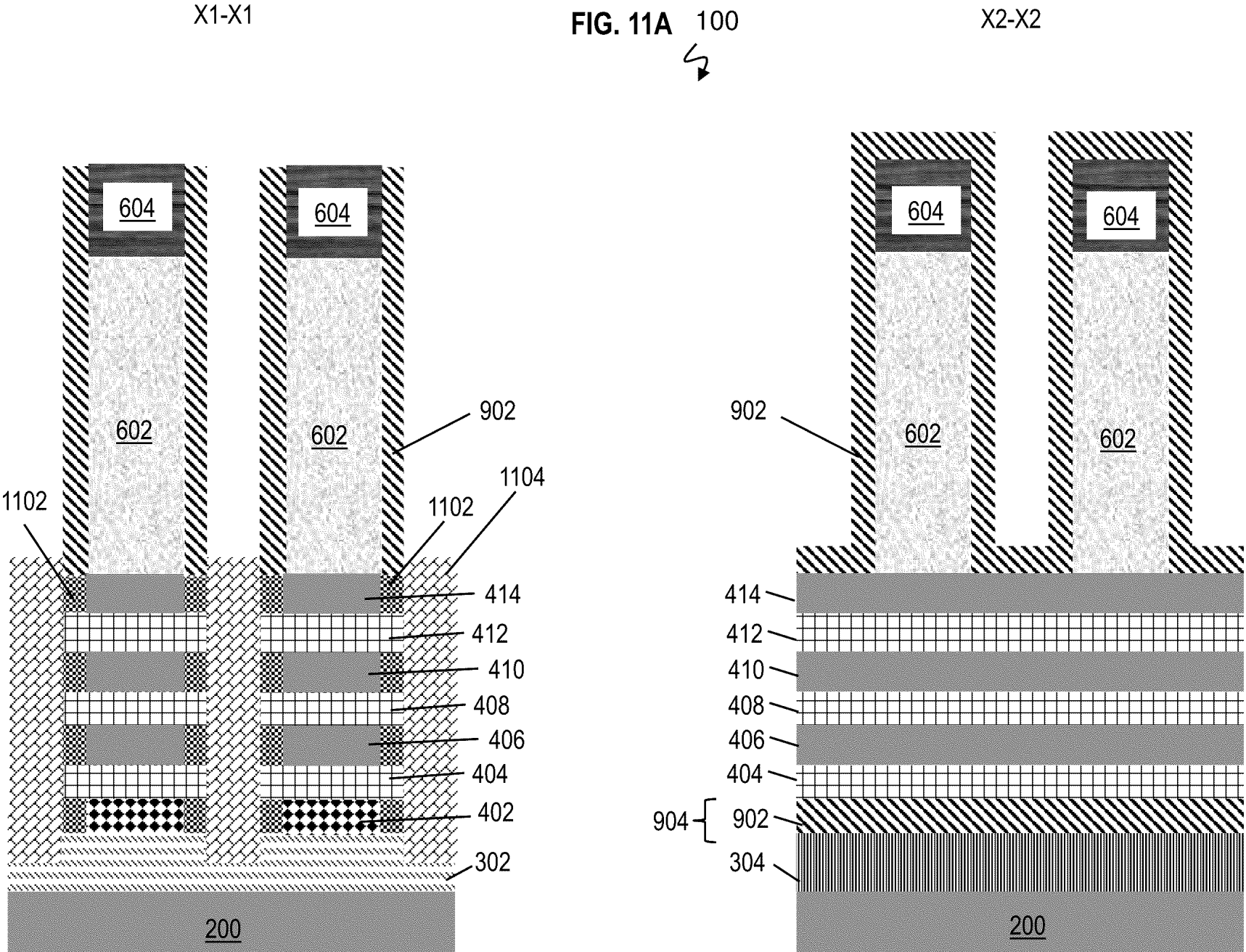


Figure 11B

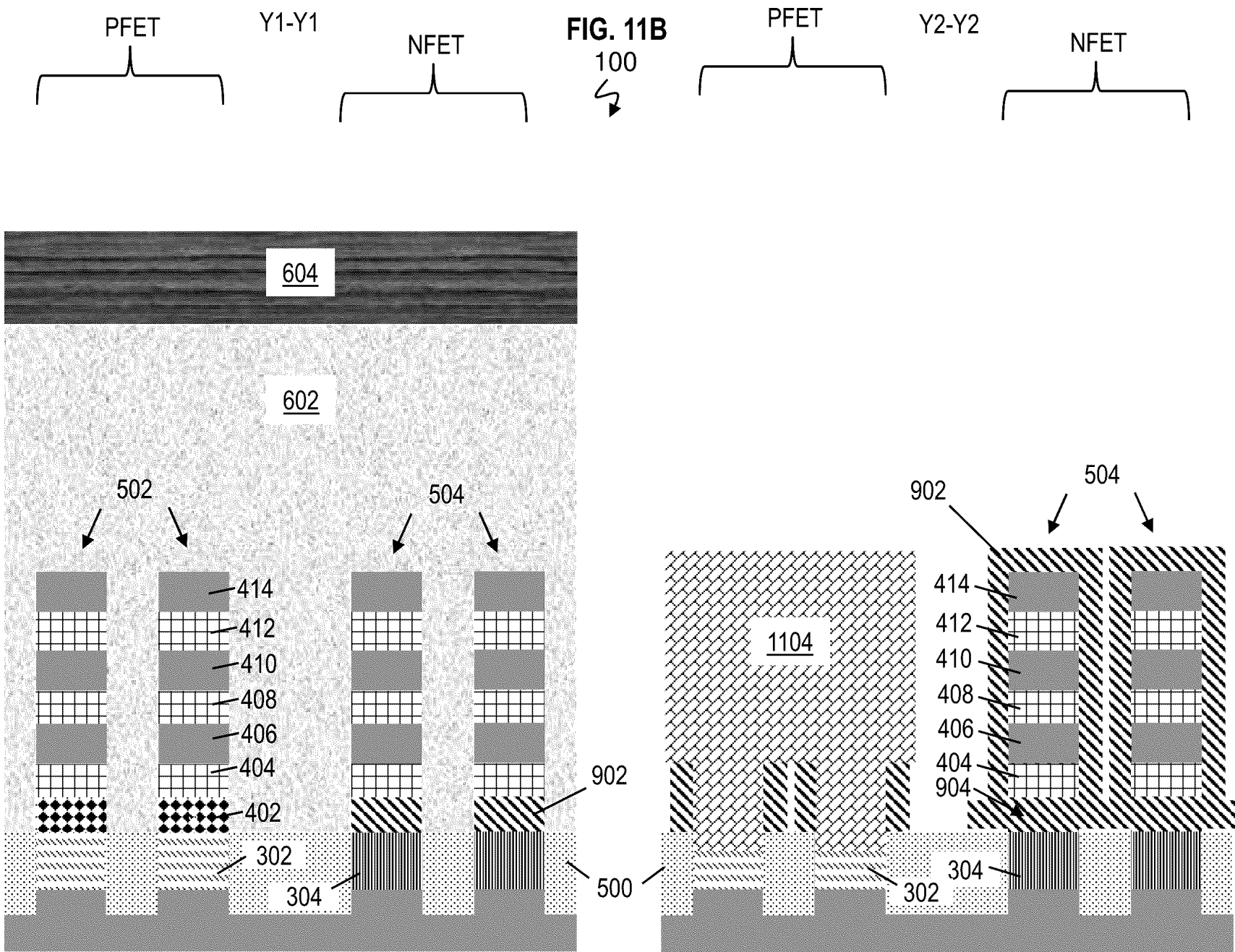


Figure 12A

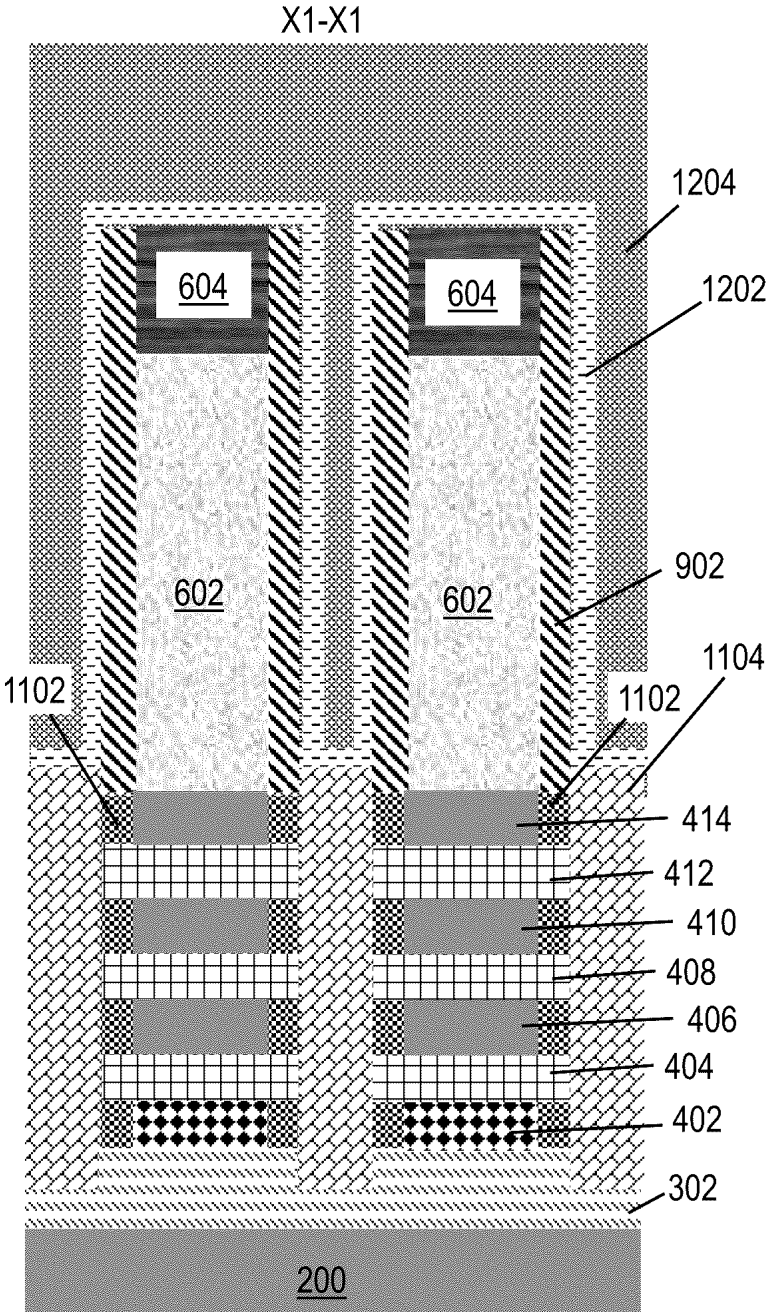


FIG. 12A 100

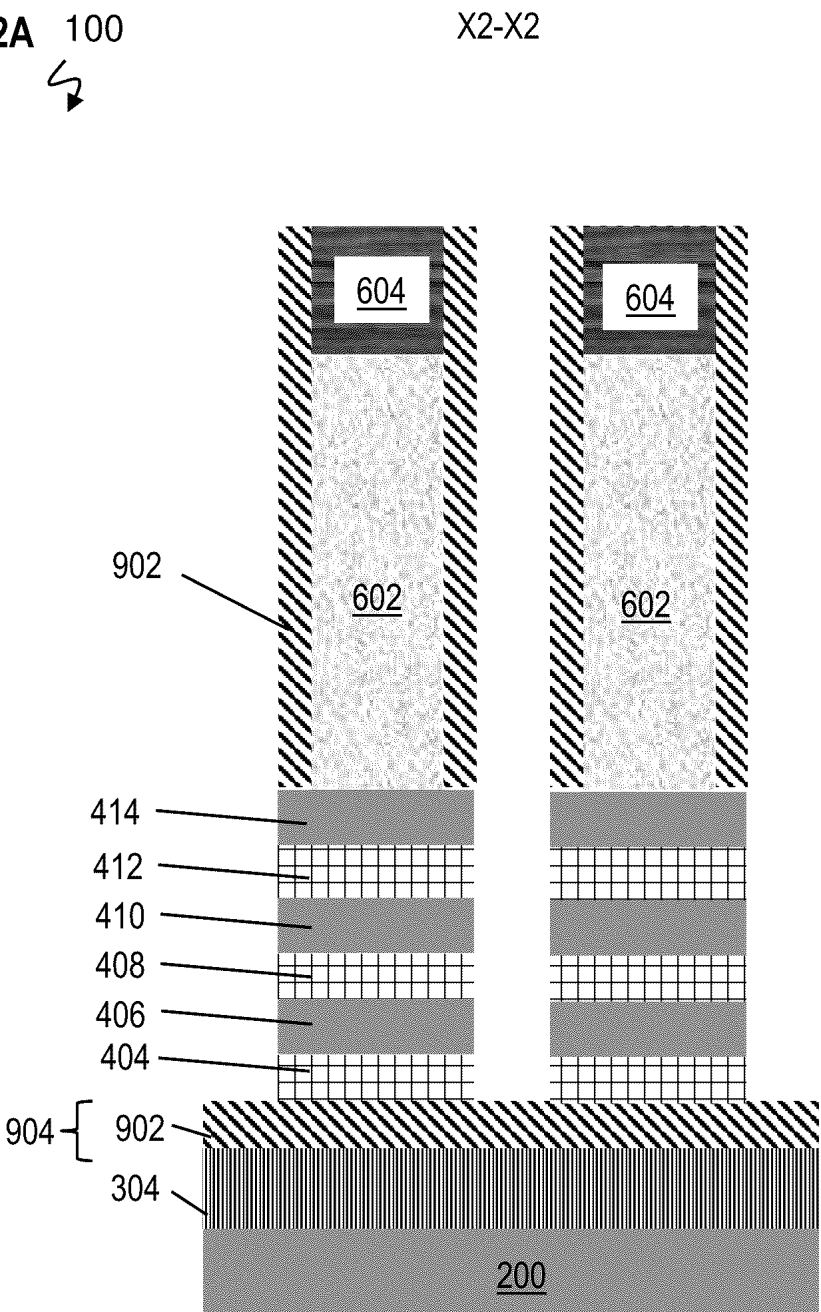


Figure 12B

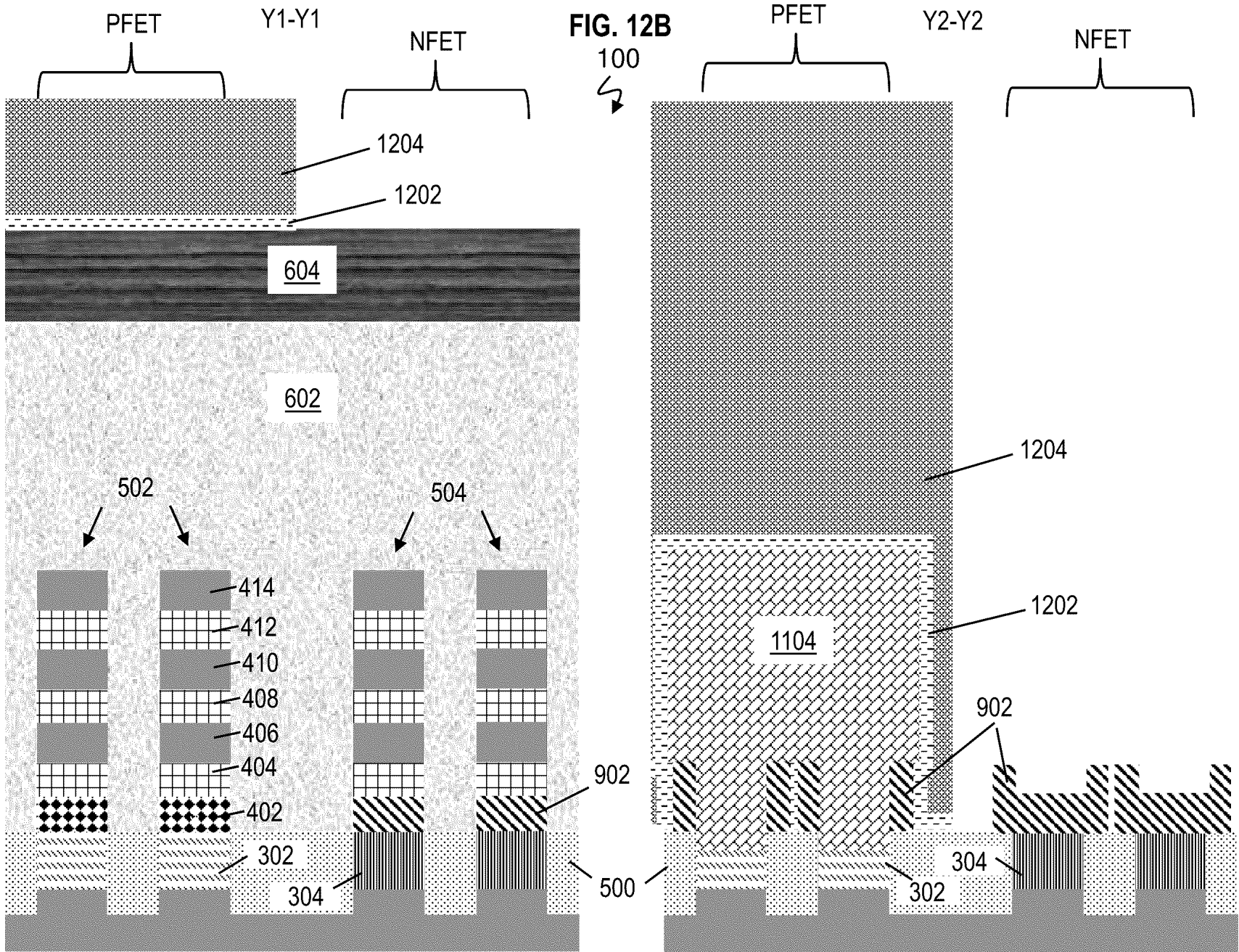


Figure 13A

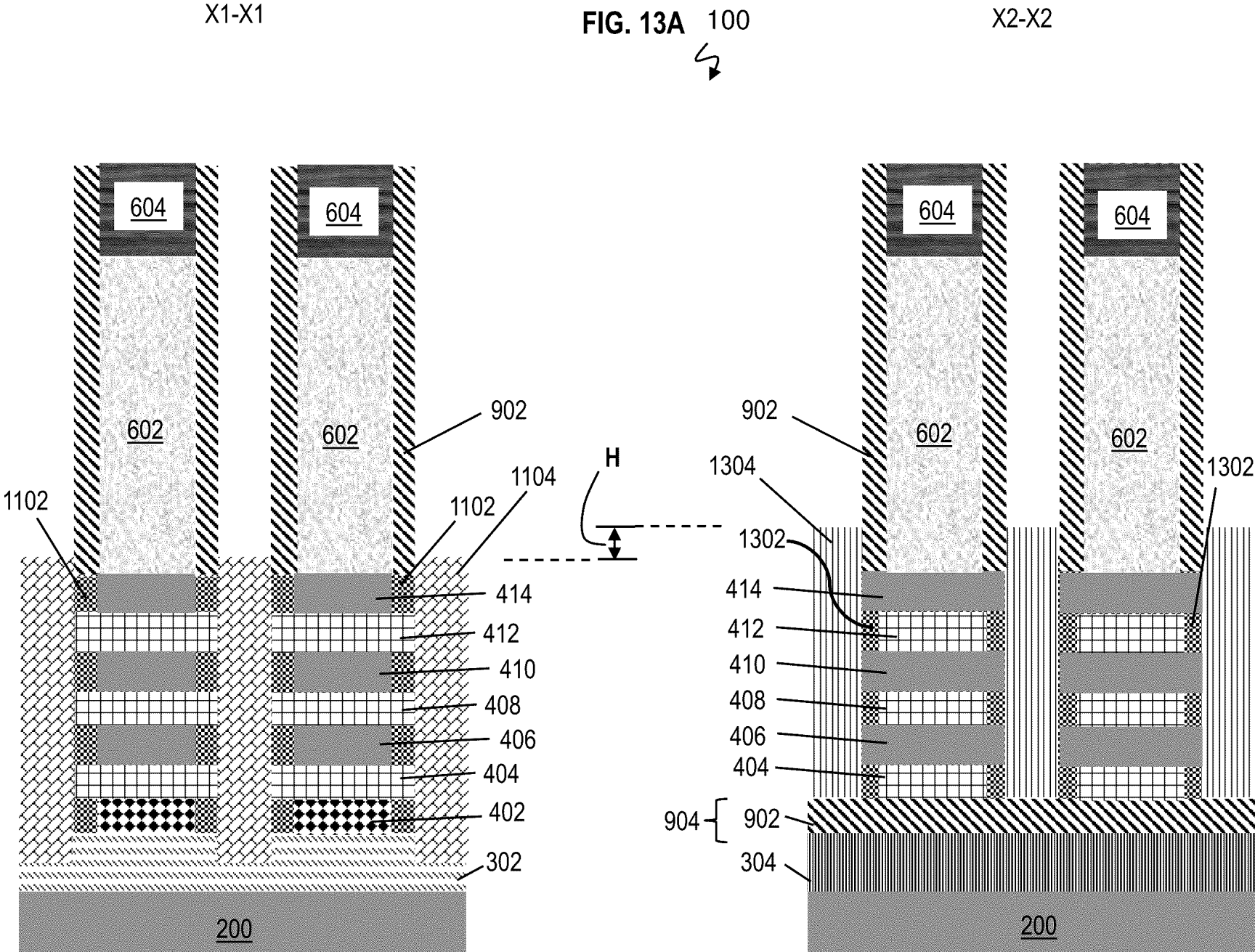
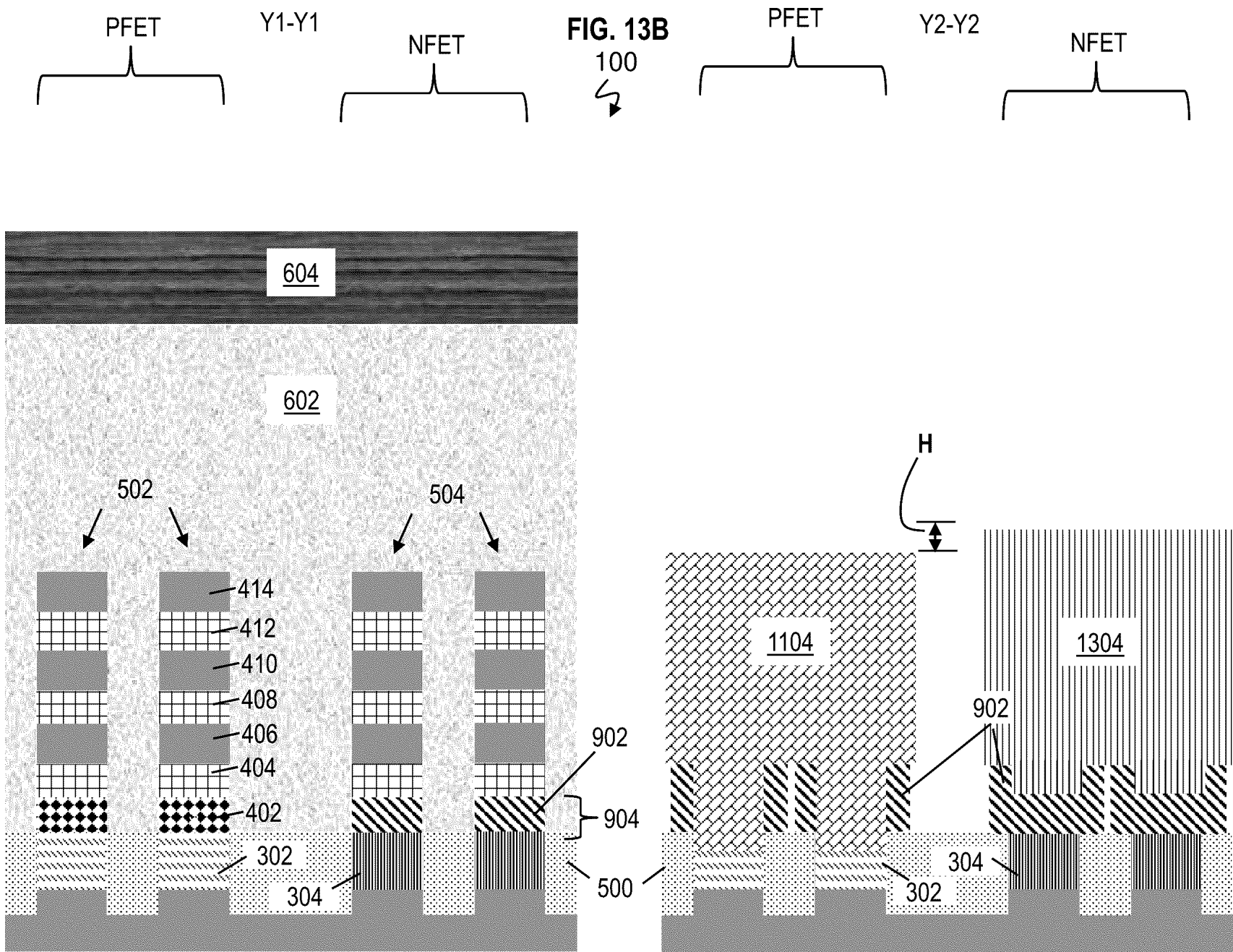


Figure 13B



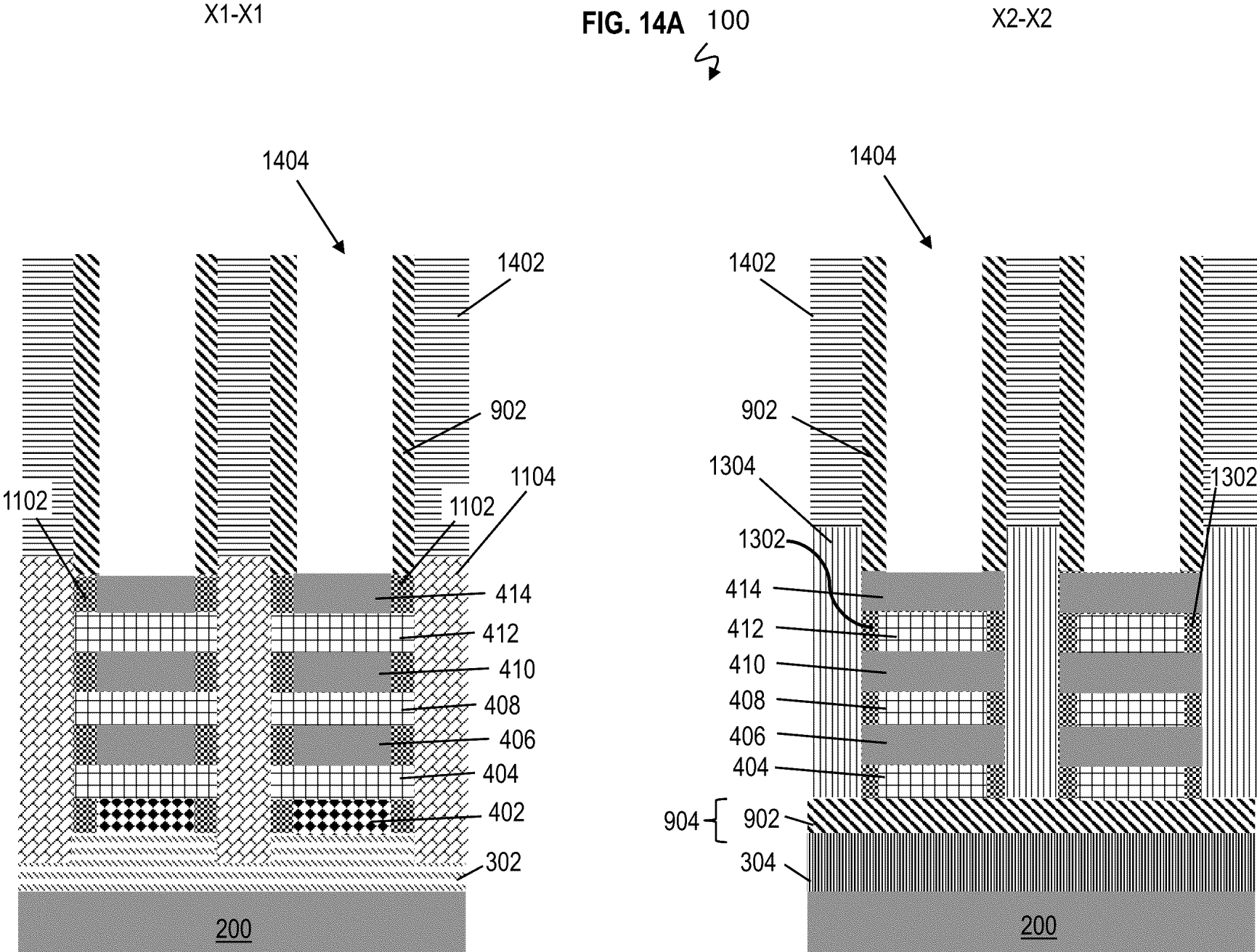
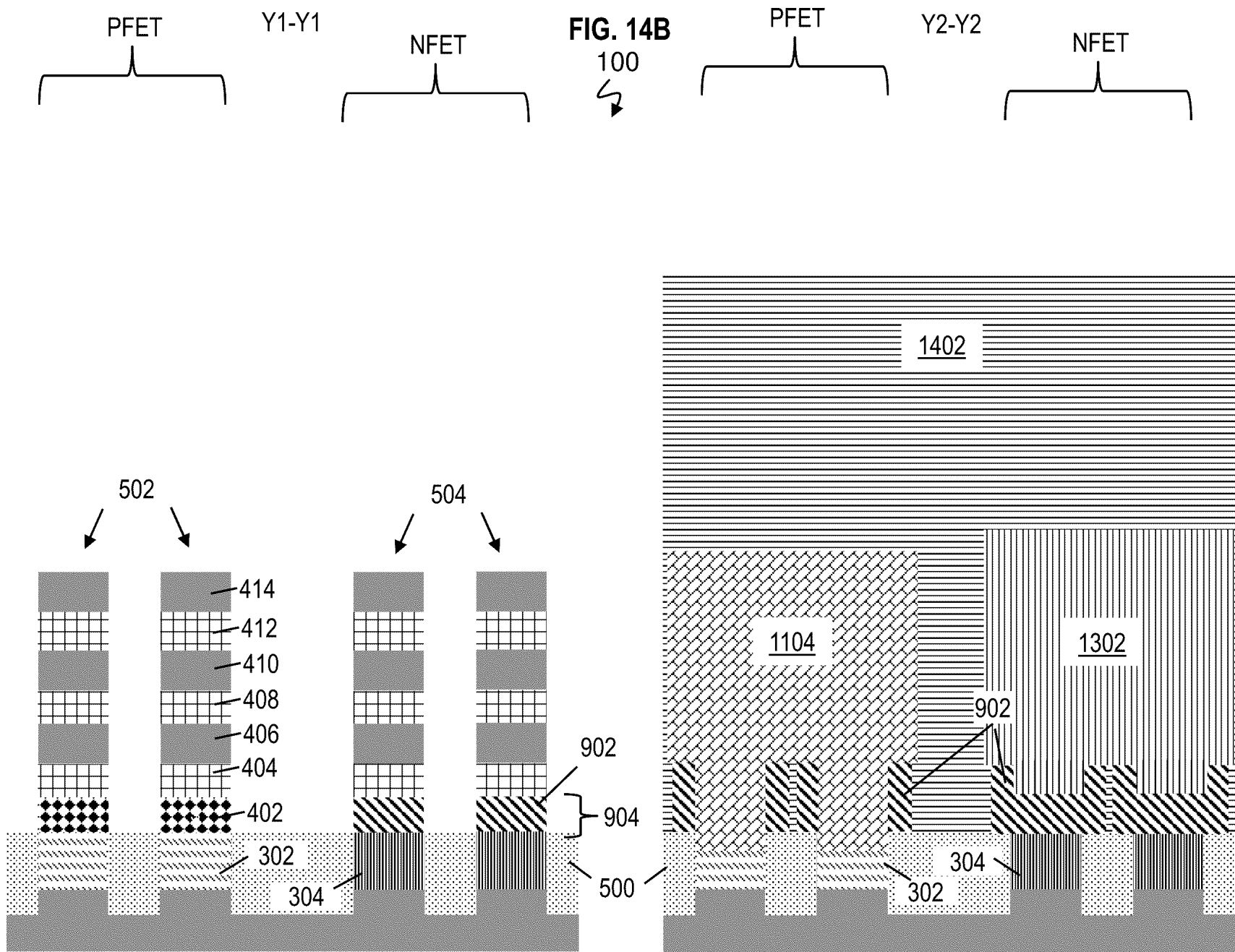


Figure 14B



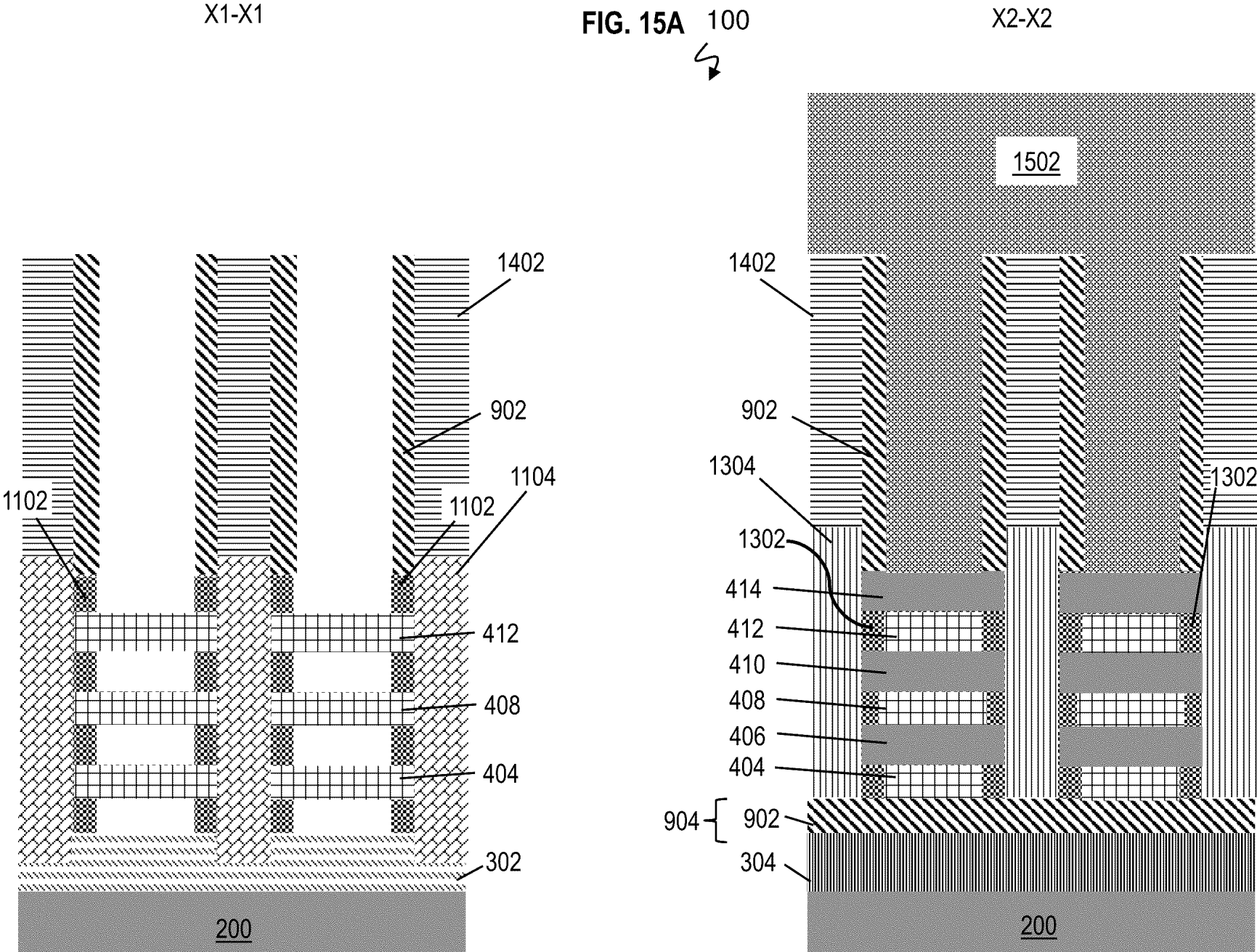
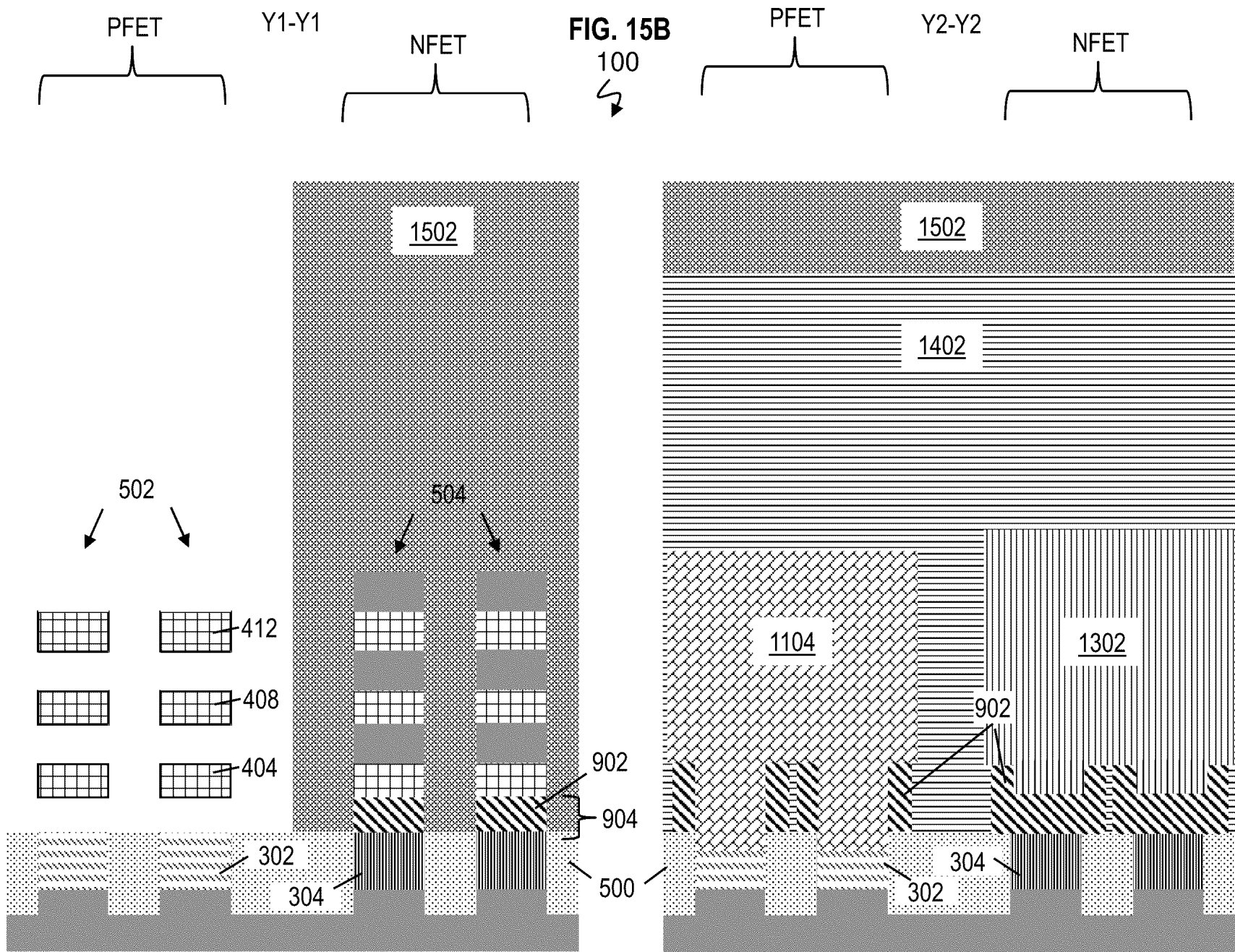


Figure 15B



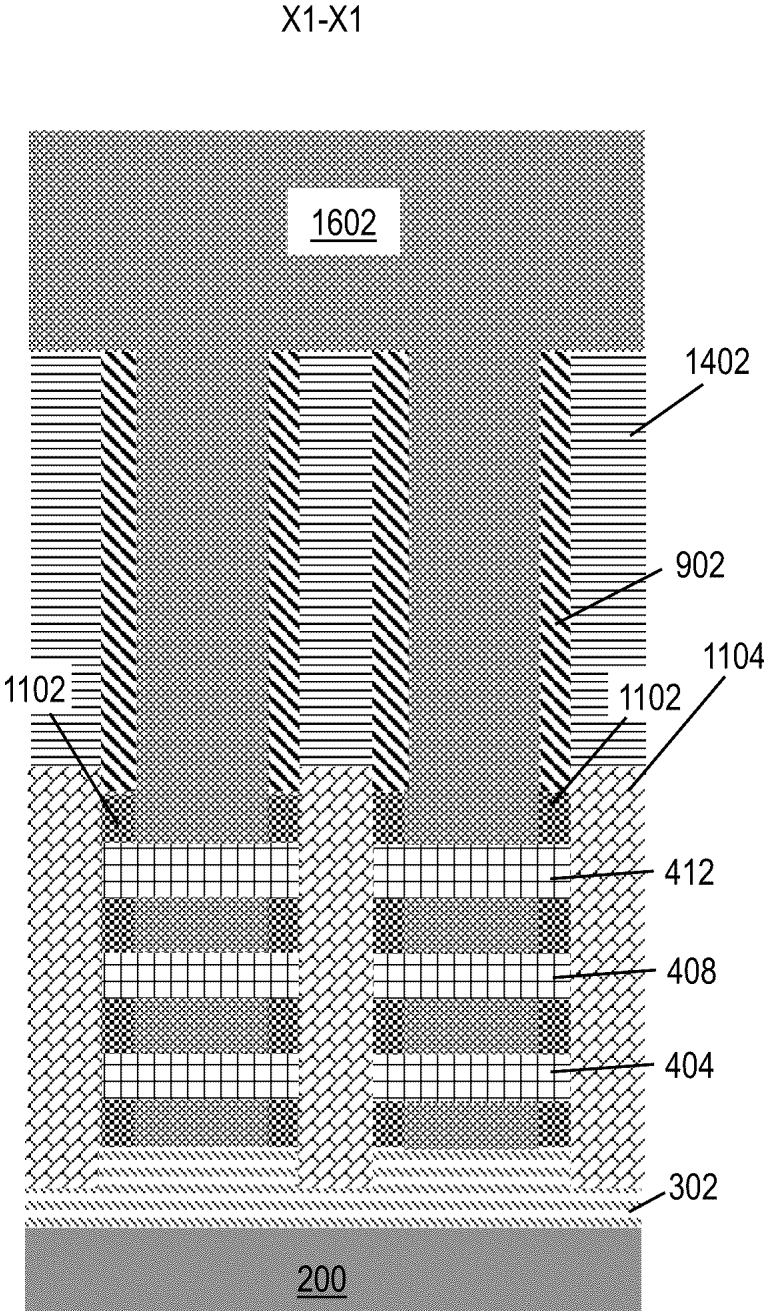


FIG. 16A 100

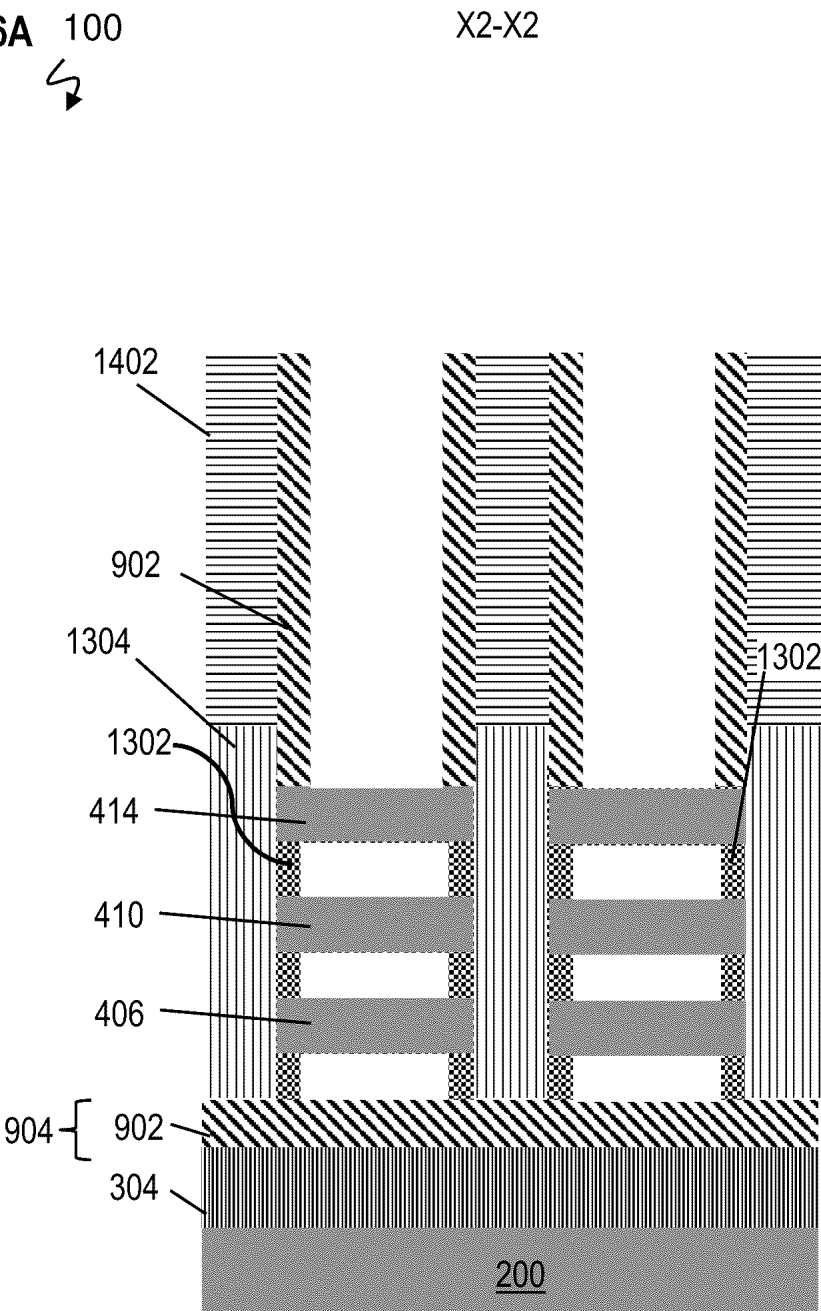
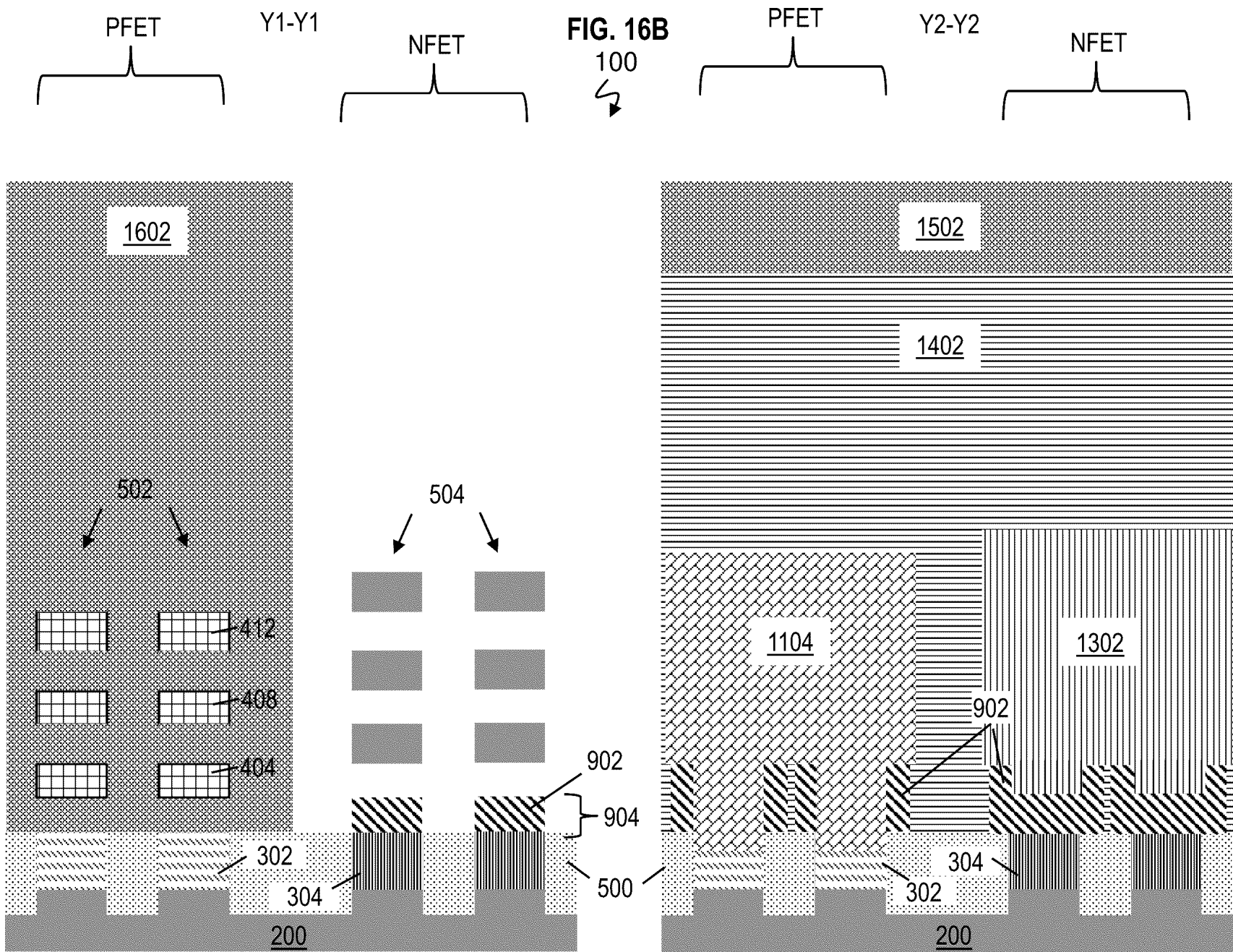


Figure 16B



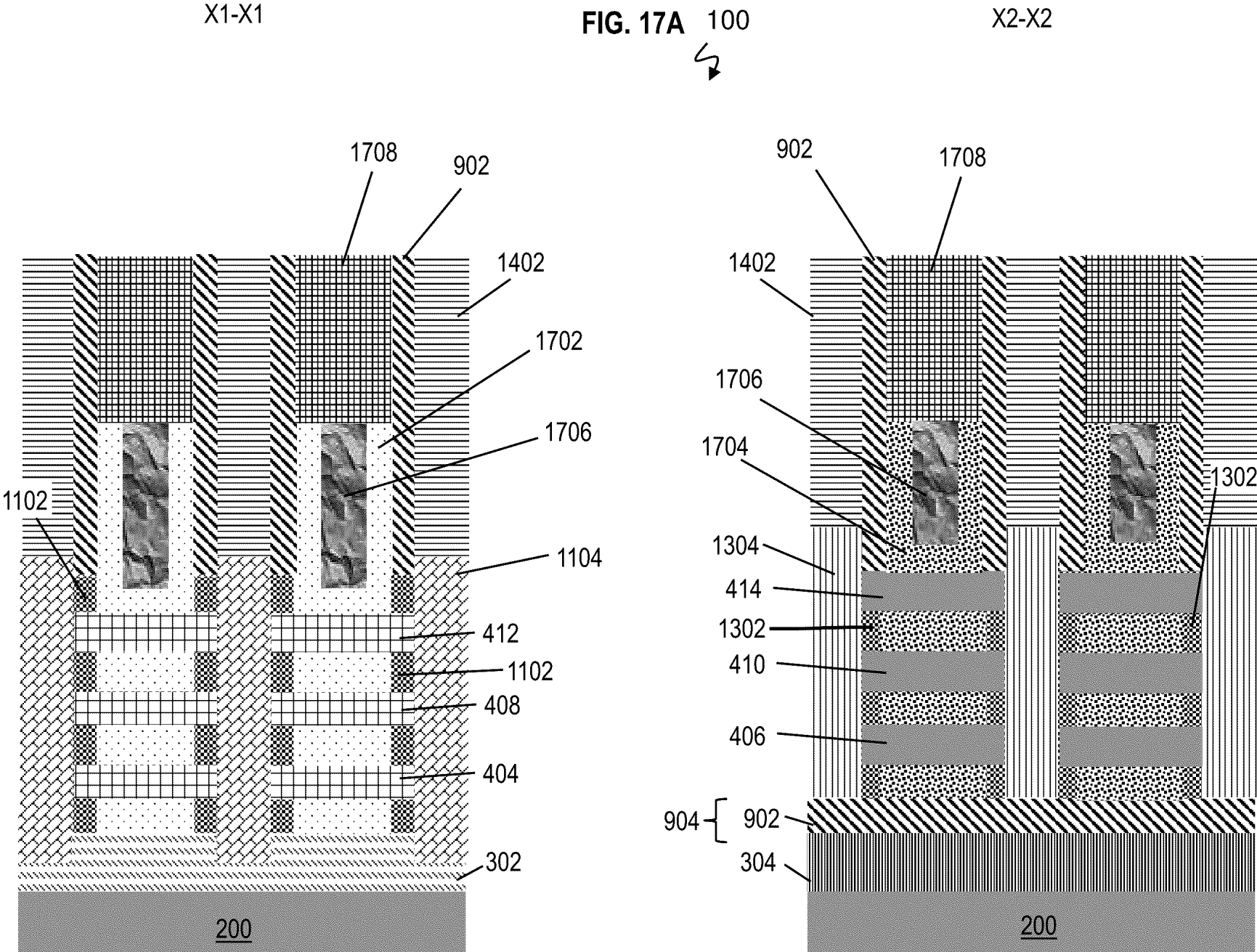


Figure 17B

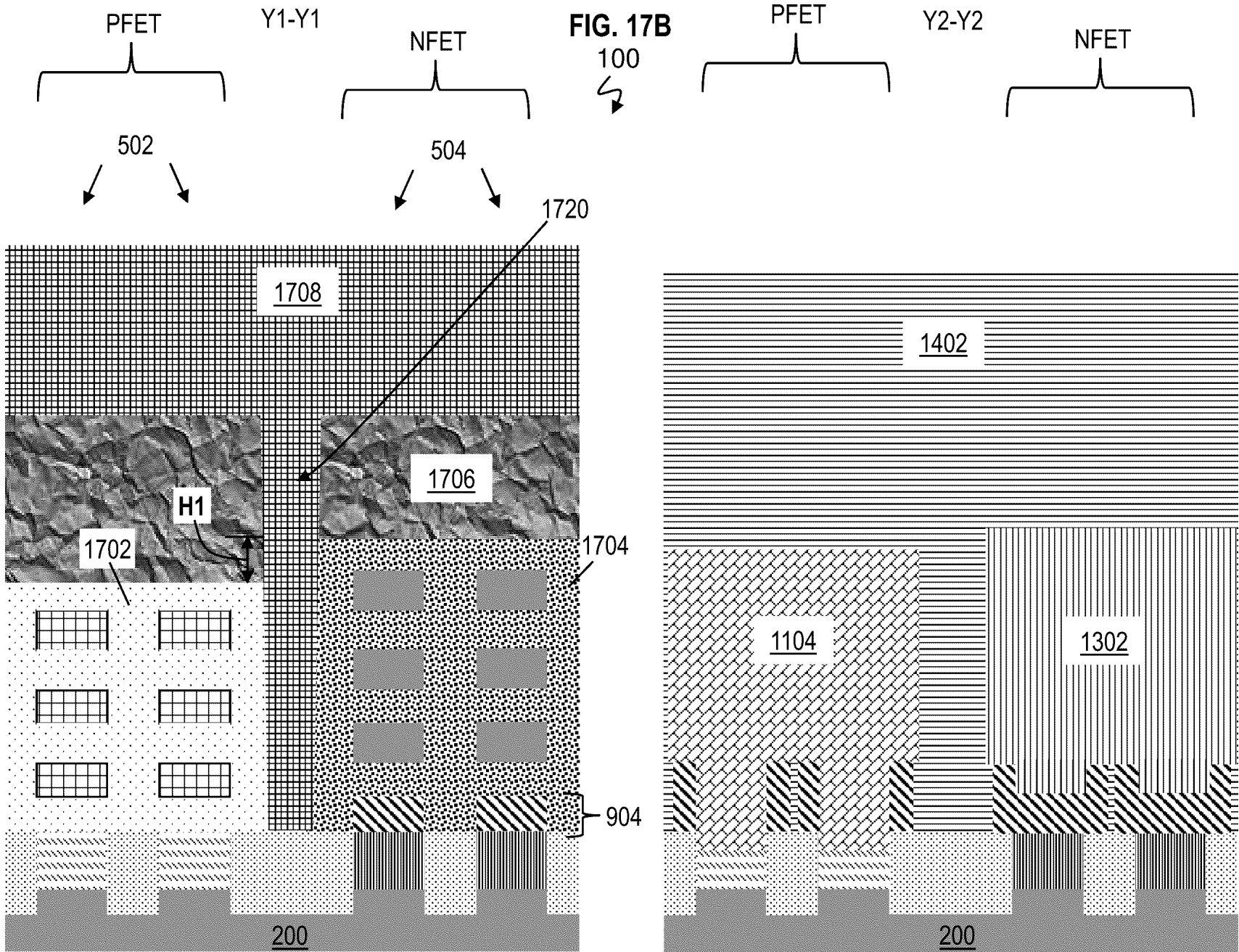


Figure 18A

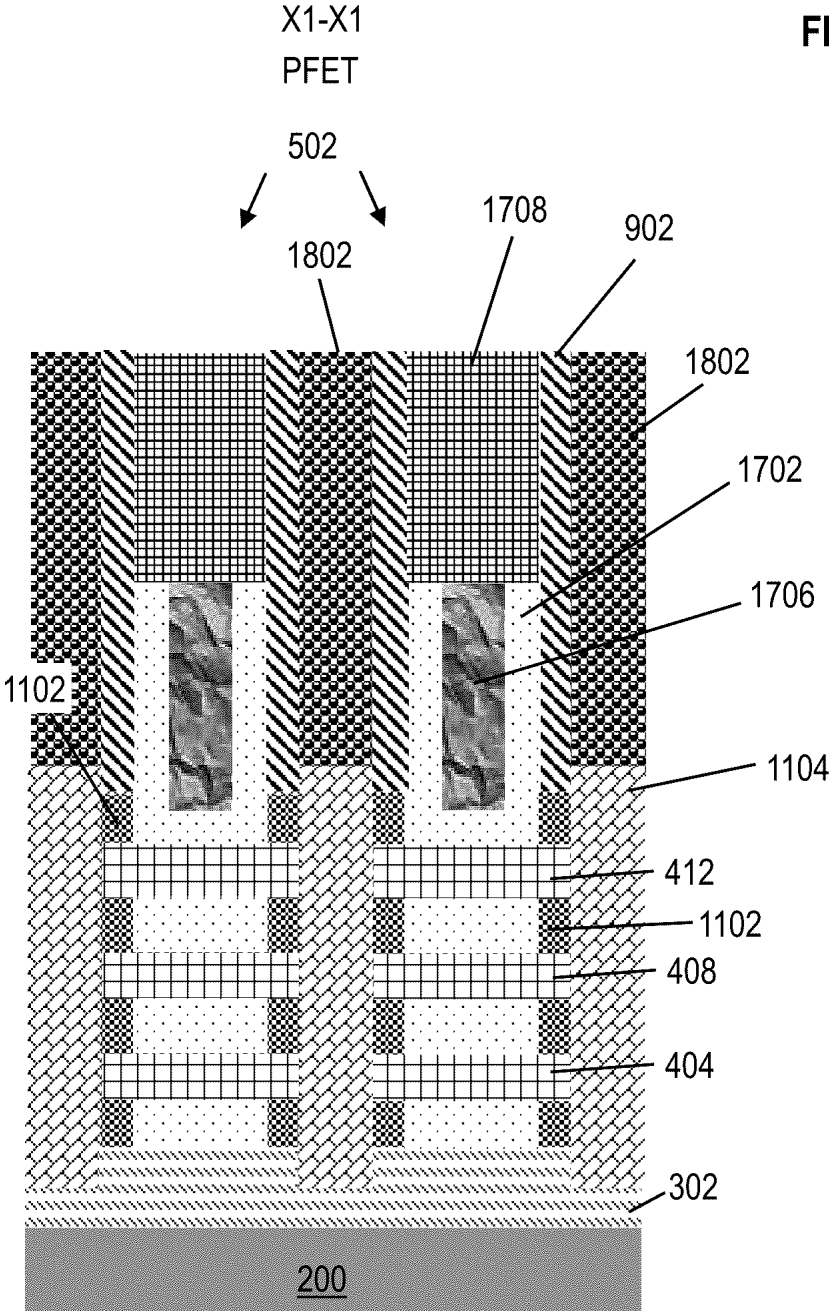


FIG. 18A 100

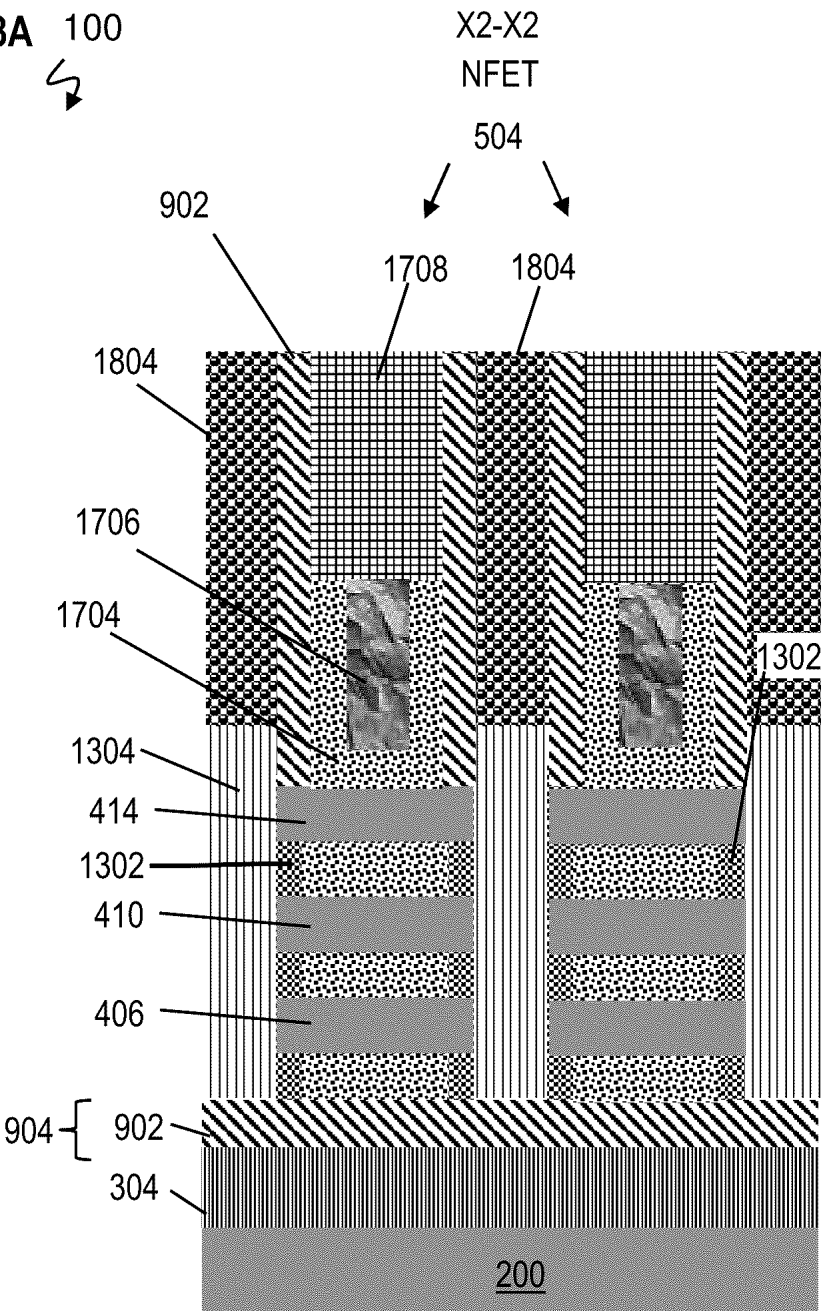


Figure 18B

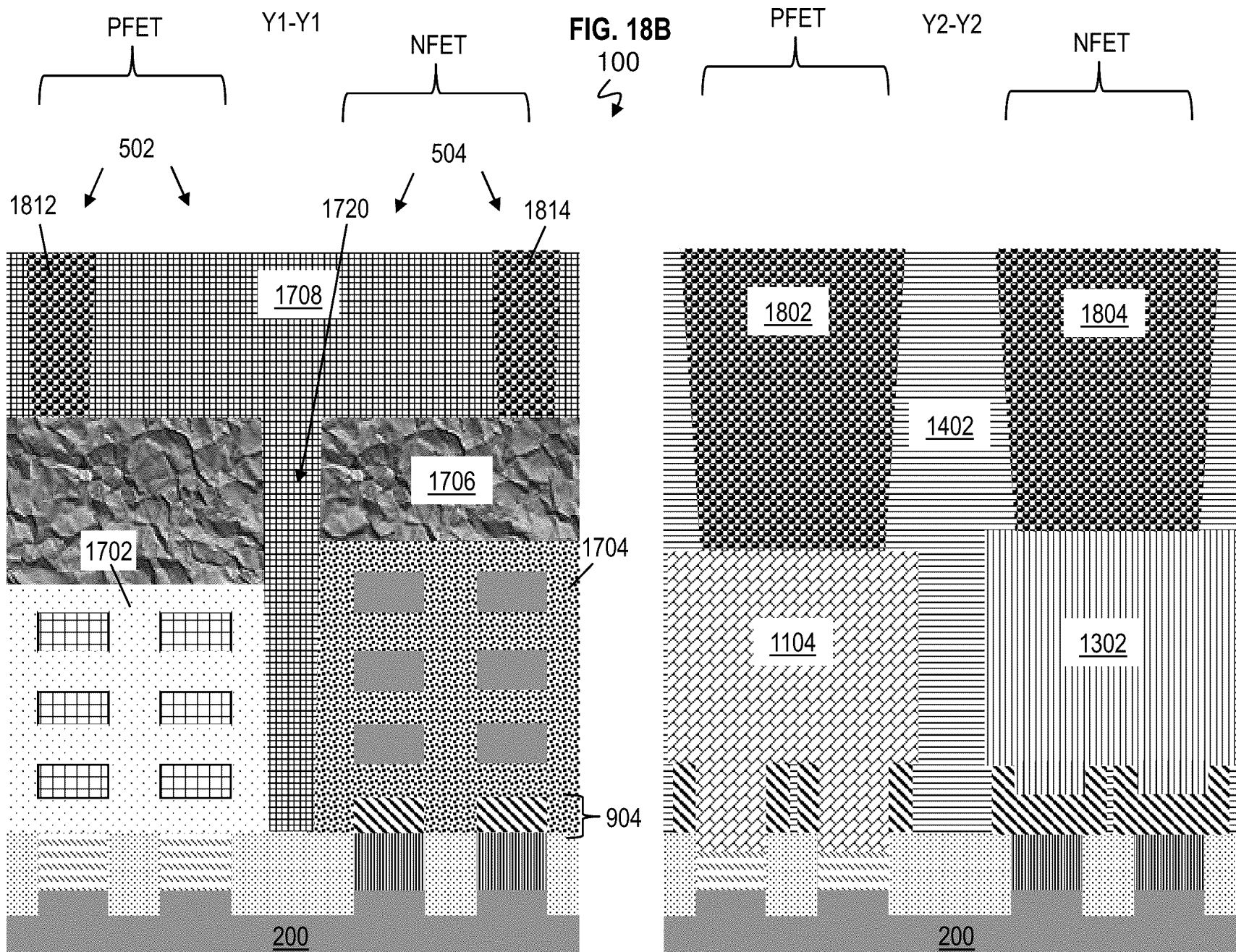


Figure 19A

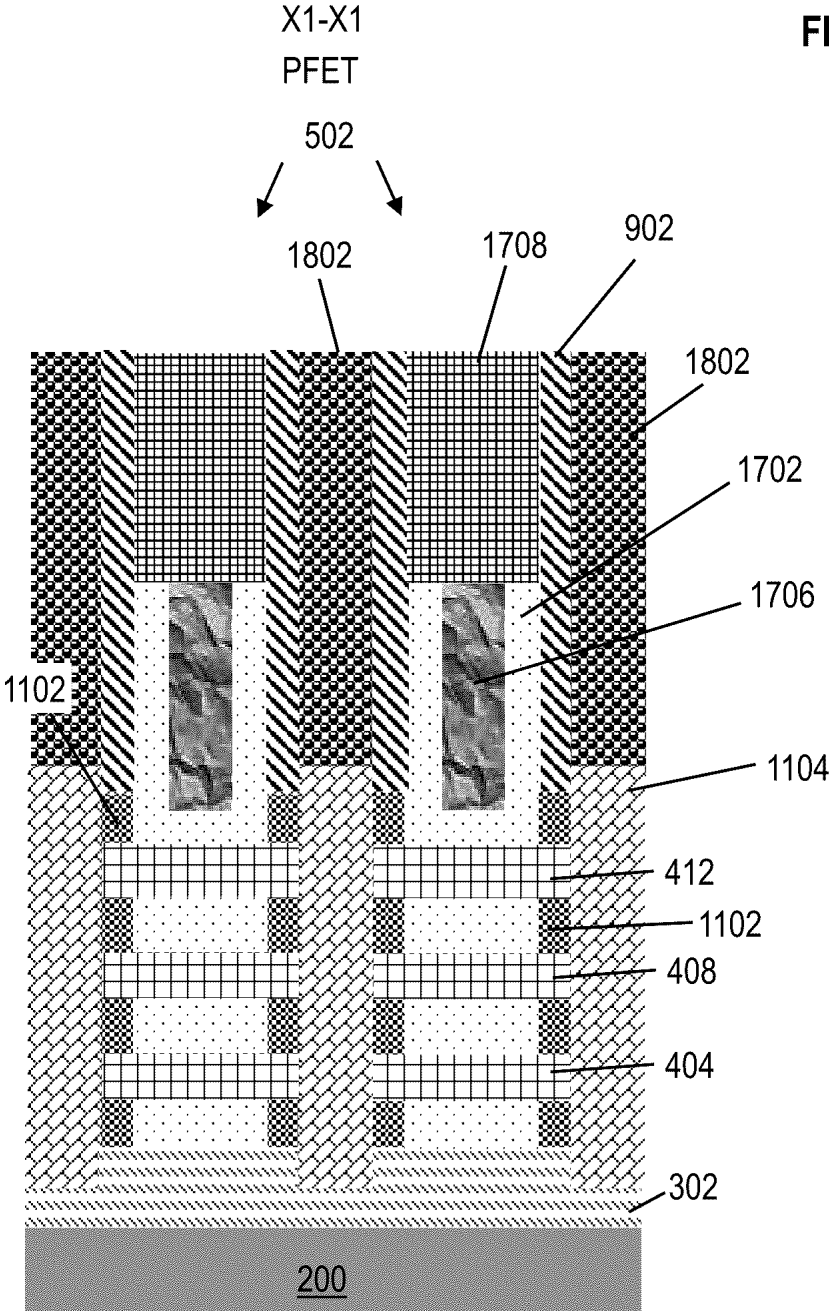


FIG. 19A 100

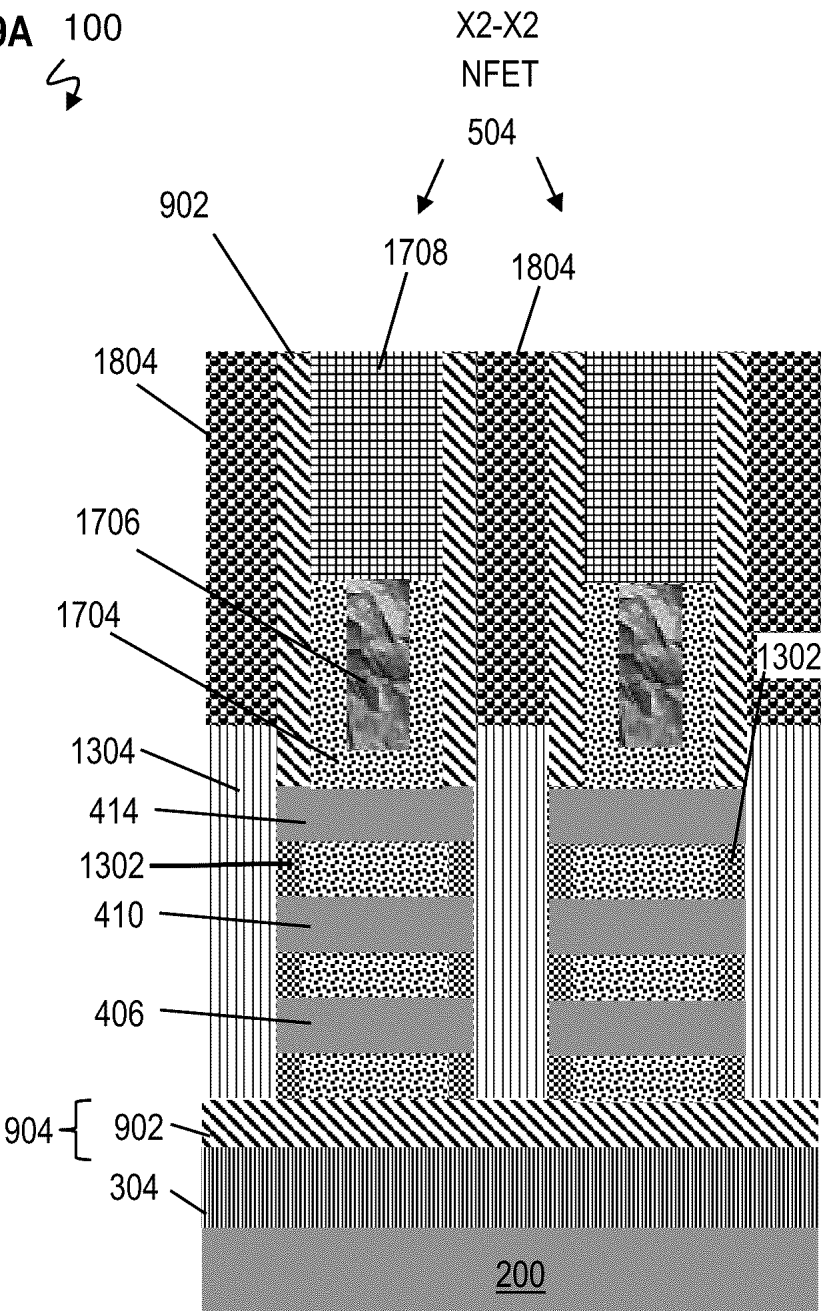


Figure 19B

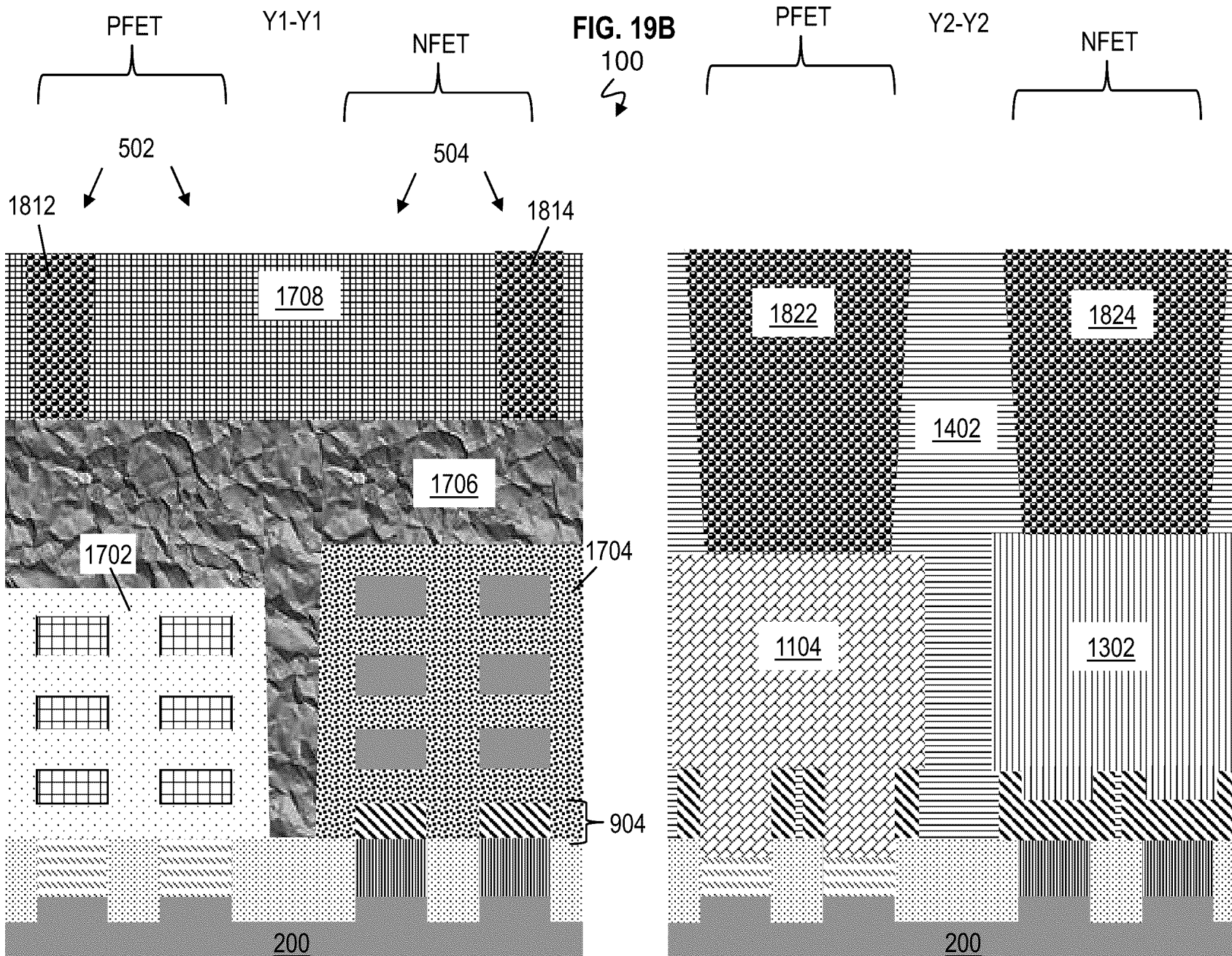


FIG. 20 2000



FORM A PFET COMPRISING AT LEAST ONE SILICON GERMANIUM CHANNEL 2002



FORM AN NFET COMPRISING AT LEAST ONE SILICON CHANNEL, THE PFET BEING POSITIONED Laterally TO THE NFET, THE AT LEAST ONE SILICON CHANNEL AND THE AT LEAST ONE SILICON GERMANIUM CHANNEL BEING STAGGERED IN A VERTICAL DIRECTION 2004

FIG. 21 2100


FORM A FIRST NANOSHEET STACK AND A SECOND NANOSHEET STACK, EACH OF THE FIRST AND SECOND NANOSHEET STACKS COMPRISING A FIRST LAYER, SECOND LAYERS, AND THIRD LAYERS 2102



REPLACE THE FIRST LAYER IN THE SECOND NANOSHEET STACK WITH A BOTTOM ISOLATION LAYER 2104



REPLACE THE FIRST LAYER AND THE THIRD LAYERS IN THE FIRST NANOSHEET STACK
WITH FIRST GATE STACK MATERIAL 2106



REPLACE THE SECOND LAYERS IN THE SECOND NANOSHEET STACK WITH SECOND
GATE STACK MATERIAL 2108

